Course Code: CCE – 3503
Course Title: Information Theory & Error Correction Coding

Prepared By-

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Course Objectives:

This course aims to address the efficient error free and secure delivery of information using binary data streams. For efficiency, the information source is coded to reduce redundancy. To minimize the effects of errors, channel coding is employed and, finally, cryptographic techniques are required to make the data secure. The aim is to present the basic theory and objectives of each of these steps, together with the basics of information theory.

Course Outcomes:

COs	**Description**
CO: 1	Familiarization with the various fields/trends of communications like Information theory, error detection and error correction related to radio and telecommunication.
CO: 2	Analysis different types of error control code properties for secure and error free efficient communications.

Course Contents:
Mid-term Exam: 30 Marks

- Information Sources and Sources Coding, Logarithmic measure for information, self and average information. Entropy, information rate, discrete sources, extensions of a discrete source, Shannon's source coding theorem.

- Markov source. Joint and conditional entropy. Source coding theorem and algorithms. Kraft inequality, Huffman code, prefix code,

- Lempel-Ziv code, rate distortion theory. Scalar and vector quantization, waveform coding. Orthogonal Codes, biorthogonal Codes. Simplex Code. Application examples: MPEG audio compression, JPEG image compression standard and MPEG video compression.

Introduction

- **Information theory** studies the transmission, processing, extraction, and utilization of information.
- Abstractly, information can be thought of as the resolution of uncertainty.

- In the case of communication of information over a noisy channel, this abstract concept was formalized in 1948 by Claude Shannon in a paper entitled **A Mathematical Theory of Communication**, in which information is thought of as a set of possible messages, and the goal is to send these messages over a noisy channel, and to have the receiver reconstruct the message with low probability of error, in spite of the channel noise.

- Shannon's main result, the noisy-channel coding theorem showed that, in the limit of many channel uses, the rate of information that is asymptotically achievable is equal to the channel capacity, a quantity dependent merely on the statistics of the channel over which the messages are sent.

- **Coding theory** is concerned with finding explicit methods, called codes, for increasing the efficiency and reducing the error rate of data communication over noisy channels to near the channel capacity.

- These codes can be roughly subdivided into
  - data compression (source coding)
  - error-correction (channel coding) techniques.

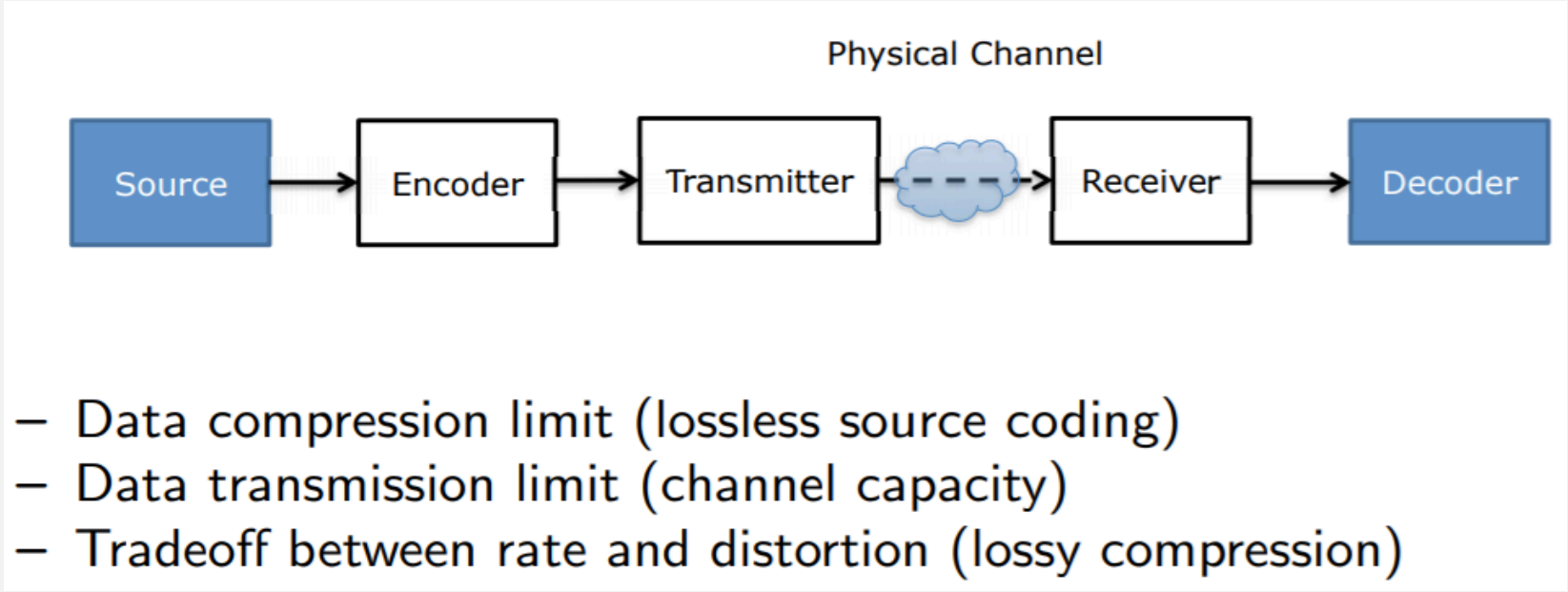
Information is the source of a communication system, whether it is analog or digital. **Information theory** is a mathematical approach to the study of coding of information along with the quantification, storage, and communication of information.

Conditions of Occurrence of Events:

If we consider an event, there are three conditions of occurrence.

- ✔ If the event has not occurred, there is a condition of **uncertainty**.
- ✔ If the event has just occurred, there is a condition of **surprise**.
- ✔ If the event has occurred, a time back, there is a condition of having some **information**.

Hence, these three occur at different times. The difference in these conditions, help us have a knowledge on the probabilities of occurrence of events.



Entropy of an Information Source

When we observe the possibilities of occurrence of an event, whether how surprise or uncertain it would be, it means that we are trying to have an idea on the average content of the information from the source of the event.

Entropy can be defined as a measure of the average information content per source symbol. Based on the probability mass function of each source symbol to be communicated, the Shannon entropy H, in units of bits (per symbol), is given by:

$$H = -\sum_i p_i \log_2 (p_i)$$

where p_i is the probability of occurrence of the i-th possible value of the source symbol. This equation gives the entropy in the units of "bits" (per symbol) because it uses a logarithm of base 2, and this base-2 measure of entropy has sometimes been called the shannon in his honor. Entropy is also commonly computed using the natural logarithm (base e, where e is Euler's number), which produces a measurement of entropy in nats per symbol and sometimes simplifies the analysis by avoiding the need to include extra constants in the formulas.

The amount of uncertainty remaining about the channel input after observing the channel output, is called as Conditional Entropy.

Discrete Memoryless Source

A source from which the data is being emitted at successive intervals, which is independent of previous values, can be termed as discrete memoryless source.

This source is discrete as it is not considered for a continuous time interval, but at discrete time intervals. This source is memoryless as it is fresh at each instant of time, without considering the previous values.

Source Coding:

The source encoder converts information waveforms to bits, while the decoder converts bits back to waveforms. Source encoding aims to convert information waveforms (text, audio, image, video, etc.) into bits, the universal currency of information in the digital world.

The three major steps are:

Sampling: Convert the continuous-time analog waveform to discrete-time sequence (but still continuous-valued).

Quantization: Convert each continuous-valued symbol to discrete-valued representatives.

Data Compression: Remove the redundancy in the data and generate roughly i.i.d. uniformly distributed bits.

Source decoding does the reverse of encoding.

In this course, we focus on the channel coding part and will not go into details of source coding.

In simpler words, the code-word (For example: Morse code for the word QUEUE is -.- ..- . ..- .) is always greater than or equal to the source code (QUEUE in example). Which means, the symbols in the code word are greater than or equal to the alphabets in the source code.

Channel Coding

The channel coding in a communication system, introduces redundancy with a control, so as to improve the reliability of the system. Source coding reduces redundancy to improve the efficiency of the system.

Channel coding consists of two parts of action.

- **Mapping** incoming data sequence into a channel input sequence.
- **Inverse mapping** the channel output sequence into an output data sequence.

The final target is that the overall effect of the channel noise should be minimized.
The mapping is done by the transmitter, with the help of an encoder, whereas the inverse mapping is done at the receiver by a decoder.

Channel Coding: The channel encoder converts bits to signal waveform, while the decoder converts received waveform back to bits.
Channel encoding aims to convert information bits into passband wave forms bits, the universal currency of information in the digital world.

The four major steps are:

- Error correcting codes: introduce redundancy into the information bits and produce longer coded bits.

  Remark Example of "error correction":

  - Repetition code: each bit repeat N times
    Channel noise: flip the bit n.p. p, w.p.$(1-p)$ remain the same.

  - Bit Error happens when there are more than $\frac{N}{2}$ bit flips

$$\Pr\{\text{error}\} = 1 - \sum_{i=0}^{\lfloor \frac{N}{2} \rfloor} \binom{N}{i} p^i (1-p)^{n-i}$$

- Symbol mapping: map the coded bits to *constellation points*, each of which is a complex symbol.
- Pulse shaping: modulate the symbol to suitable *baseband* waveforms. There are some specific conditions needed to be satisfied, which will be discussed in later lectures.
- Up conversion: convert the baseband waveform to *passband* waveform, so that the effective frequency band follows the constraints from the physical world.

Channel decoding does the reverse of encoding.

LECTURE – 3

Measure the Information

- To measure the information content of a message quantitatively, we are required to arrive at an intuitive concept of the amount of information.
- **Information Measure** - a system of measurement of information based on the probabilities of the events that convey information.

Logarithmic Measure for Information

- The mathematical theory of information is based on probability theory and statistics, and measures information with several quantities of information.
- The choice of logarithmic base in the following formulae determines the unit of information entropy that is used.

- The most common unit of information is the bit, based on the binary logarithm.
- Other units include the nat, based on the natural logarithm, and the hartley, based on the base 10 or common logarithm.

Unit of Information Measure

Base of the logarithm will determine the unit assigned to the information content.

Natural logarithm base : nat

Base - 10 : Hartley / decit

Base - 2 : bit

Use of binary digit as the unit of information?

Is based on the fact that if two possible binary digits occur with equal probability, (p1 = p2 = ½) then the correct identification of the binary digit conveys an amount of information.

$$I(m1) = I(m2) = -\log_2(½) = 1 \text{ bit}$$

One bit is the amount if information that we gain when one of two possible and equally likely events occurs.

Illustrative Example

A source puts out one of five possible messages during each message interval. The probs. of these messages are $p_1 = \frac{1}{2}$; $p_2 = \frac{1}{4}$; $p_1 = \frac{1}{4}$: $p_1 = \frac{1}{16}$, $p_5 \frac{1}{16}$

What is the information content of these messages?

$$I(m_1) = -\log_2 \frac{1}{2} = 1 \text{ bit}$$

$$I(m_2) = -\log_2 \frac{1}{4} = 2 \text{ bits}$$

$$I(m_3) = -\log \frac{1}{8} = 3 \text{ bits}$$

$$I(m_4) = -\log_2 \frac{1}{16} = 4 \text{ bits}$$

$$I(m_5) = -\log_2 \frac{1}{16} = 4 \text{ bits}$$

Self-information

Shannon derived a measure of information content called the self-information or "surprisal" of a message m:

$$\mathrm{I}(m) = \log\left(\frac{1}{p(m)}\right) = -\log(p(m))$$

- Where, p(m)=Pr(M=m) is the probability that message m is chosen from all possible choices in the message space M.
- The base of the logarithm only affects a scaling factor and, consequently, the units in which the measured information content is expressed.
- If the logarithm is base 2, the measure of information is expressed in units of bits.

Entropy

The **entropy** of a discrete message space M is a measure of the amount of **uncertainty** one has about which message will be chosen. It is defined as the average self-information of a message m from that message space:

$$\mathrm{H}(M) = \mathbb{E}\left[\mathrm{I}(M)\right] = \sum_{m \in M} p(m)\,\mathrm{I}(m) = -\sum_{m \in M} p(m) \log p(m).$$

where

$\mathbb{E}[-]$ denotes the expected value operation.

An important property of entropy is that it is maximized when all the messages in the message space are equiprobable (e.g. $p(m) = 1/|M|$). In this case $\mathrm{H}(M) = \log |M|$.

Sometimes the function H is expressed in terms of the probabilities of the distribution:

$$\mathrm{H}(p_1, p_2, \ldots, p_k) = -\sum_{i=1}^{k} p_i \log p_i, \text{ where each } p_i \geq 0 \text{ and } \sum_{i=1}^{k} p_i = 1.$$

An important special case of this is the **binary entropy function**:

$$\mathrm{H_b}(p) = \mathrm{H}(p, 1-p) = -p \log p - (1-p) \log(1-p).$$

Average Information Content of Symbols in Long Independence Sequences

Suppose that a source is emitting one of M possible symbols s0, s1 ….. s M in a statically independent sequence.

Let p1, p2,…….. p M be the problems of occurrence of the M-symbols respectively.

Suppose further that during a long period of transmission a sequence of N symbols have been generated.

On an average – S1 will occur NP1 times
S2 will occur NP2 times
Si will occur NPi times

The information content of the ith symbol is I(si) = $\log\frac{1}{P_i}$ bits

So, PiN occurrences of Si contributes an information content of i ii $\frac{1}{P_i}$ bits.

PN.I(s)=PN.log

Total information content of the message is = Sum of the contribution due to each of M symbols of the source alphabet.

M 1 i.e., $\mathbf{I_{total} = \sum NP1 \log\frac{1}{P_i}}$ **bits**

Average information content: $\boldsymbol{H = \sum_{i=1}^{M} p \log\frac{1}{p_i}\ bits/symbol}$

This is equation used by Shannon.
Average information content per symbol is also called the source entropy.

- **Information rate**

If the source is emitting symbols at a fixed rate of ‘’r_s’ symbols / sec, the average source information rate ‘R’ is defined as –

$$\mathbf{R = r_s \, . \, H \ \ bits / sec}$$

1. Consider a discrete memoryless source with a source alphabet A = { s_0, s_1, s_2} with respective probs. $p_0 = ¼$, $p_1 = ¼$, $p_2 = ½$. Find the entropy of the source.

Solution: By definition, the entropy of a source is given by

$$H = \sum_{i=1}^{M} p_i \log \frac{1}{p_i} \text{ bits/ symbol}$$

H for this example is

$$H(A) = \sum_{i=0}^{2} p_i \log \frac{1}{p_i}$$

Substituting the values given, we get

$$H(A) = p_o \log \frac{1}{P_o} + P_1 \log \frac{1}{p_1} + p_2 \log \frac{1}{p_2}$$

$$= ¼ \log_2 4 + ¼ \log_2 4 + ½ \log_2 2$$

$$= \frac{3}{2} = 1.5 \text{ bits}$$

if $r_s = 1$ per sec, then

$H'(A) = r_s$ H (A) = 1.5 bits/sec

2. An analog signal is band limited to B Hz, sampled at the Nyquist rate, and the samples are quantized into 4-levels. The quantization levels Q_1, Q_2, Q_3, and Q_4 (messages) are assumed independent and occur with probs.

$P_1 = P_2 = \frac{1}{8}$ and $P_2 = P_3 = \frac{3}{8}$. Find the information rate of the source.

Solution: By definition, the average information H is given by

$$H = p_1 \log \frac{1}{p_1} + p_2 \log \frac{1}{p_2} + p_3 \log \frac{1}{p_3} + p_4 \log \frac{1}{p_4}$$

Substituting the values given, we get

$$H = \frac{1}{8} \log 8 + \frac{3}{8} \log \frac{8}{3} + \frac{3}{8} \log \frac{8}{3} + \frac{1}{8} \log 8$$

$$= 1.8 \text{ bits/ message.}$$

Information rate of the source by definition is

$$R = r_s \quad H$$

$$R = 2B, (1.8) = (3.6\ B) \text{ bits/sec}$$

3. Compute the values of H and R, if in the above example, the quantities levels are so chosen that they are equally likely to occur,

Solution:

Average information per message is

$$H = 4\ (\tfrac{1}{4} \log_2 4) = 2 \text{ bits/message}$$

and $R = r_s\ H = 2B\ (2) = (4B)$ bits/sec

A black and white TV picture consists of 525 lines of picture information. Assuming each line consists 525 picture elements, each element having 256 brightness levels and the pictures are repeated at the rate of 30 per sec. Calculate the average information converged by a TV set to a viewer.

A black and white TV picture consists of 525 lines of picture information. Assuming each line consists 525 picture elements, each element having 256 brightness levels and the pictures are repeated at the rate of 30 per sec. Calculate the average information converged by a TV set to a viewer.

Solution:

One picture contains 525 lines

One picture line contains 525 picture elements.

Pictures are sent at 30 per sec

Picture element rate (r) = $525 \times 525 \times 30$

$= 8.26875 \times 10^6$ picture elements/sec

$H = \log_2 M$

$M = 256$ brightness levels

$H = \log_2 (256)$

$H = 8$ bits

$R = 2H$

$= 8.26875 \times 10^6 \times 8$

$= 66.15$ mbps.

DISCRETE SOURCE

- A source of information whose output has an alphabet of distinct letters or, in the case of a physical source, whose output is a signal that is discrete in time and amplitude (see discrete and continuous systems).
- The size of the alphabet, or the number of amplitude levels, is usually finite, although for mathematical analysis it may conveniently be regarded as potentially infinite.
- The output of a discrete source is a sequence of symbols from a known discrete alphabet X .
- This alphabet could be the alphanumeric characters, the characters on a computer keyboard, English letters, Chinese characters, the symbols in sheet music (arranged in some systematic fashion), binary digits, etc.

ABDUCTIONS	ABDUCTORES	ABJECTIONS	ABORTIVELY	ABRIDGMENT
ABRUPTIONS	ABSOLUTIVE	ABSOLUTIZE	ABSORPTIVE	ACHONDRITE
ADENOVIRUS	ADIPOCYTES	ADJUNCTIVE	ADMIXTURES	ADMONISHER
ADOPTIVELY	ADSORPTIVE	ADVECTIONS	AFTERGLOWS	AFTERSHOCK

The **discrete memoryless source (DMS)** has the property that its output at a certain time does not depend on its output at any earlier time.

The **discrete source with memory (DSM)** has the property that its output at a certain time may depend on its outputs at a number of earlier times: if this number is finite, the source is said to be of finite order, otherwise it is of infinite order. DSMs are usually modeled by means of Markov chains; they are then called Markov sources.

- successive symbols statistically independent
- $S = \{s_1, s_2, \ldots, s_n\}$
- $\{p(s_1), p(s_2), \ldots, p(s_n)\}$
- $I(s_i)$, the information revealed by the occurrence of a certain source symbol, is defined as

$$I(s_i) = \log_2 \frac{1}{p(s_i)}$$

- Average Information per source symbol, entropy $H(s) = \Sigma\, p(s_i) I(s_i) = -\Sigma\, p(s_i) \log_2 p(s_i)$ bits/symbol

Extensions of a Discrete Memoryless Source

- DMS S with an alphabet of size n
- the output of the source grouped into blocks of N symbols
- S^N with an alphabet of size n^N: the Nth extension of the source S
- For a memoryless source, the probability of a symbol $\sigma_i = (s_{i_1}, s_{i_2}, \ldots, s_{i_N})$ from S^N is given by

$$p(\sigma_i) = p(s_{i_1})p(s_{i_2}) \ldots p(s_{i_N})$$

$$H(S^N) = NH(S)$$

LECTURE – 4

SHANNON'S SOURCE CODING THEOREM

- ✔ The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point (Claude Shannon, 1948)
- ✔ Why Shannon used the word reproducing and not receiving?
- ✔ What is meant by exactly and approximately

SmOkInG CaN KiLL YoU!

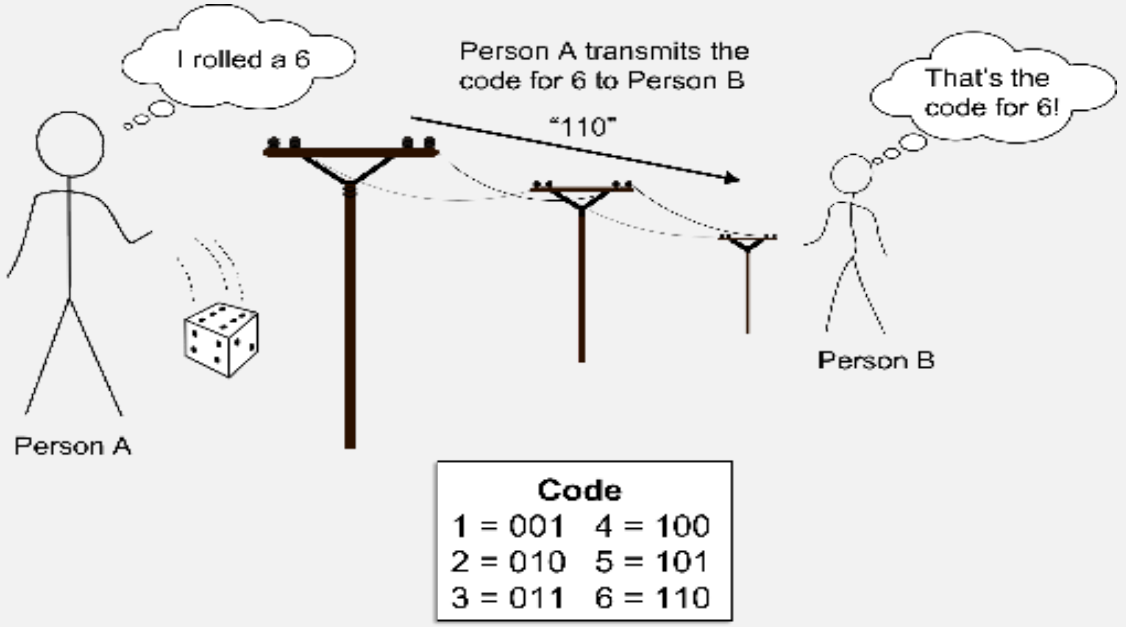
- ✔ How can we achieve perfect communication over an imperfect, noisy communication channel?
- ✔ Example: an analogue telephone line, over which two modems communicate digital Information.

modem → line+noise → modem

"THEN, OUR PROBLEM IS THE NOISE!"

- ✔ The theory provides answers to two fundamental questions (among others):What is the irreducible complexity below which a signal cannot be compressed? (Shannon Theorem 1)
- ✔ How Can we correct errors or know their locations (Shannon Theorem 2)
- ✔ What is the ultimate transmission rate for reliable communication over a noisy channel? (Shannon Theorem 3)
- ✔ How can we protect them against hacking? (Shannon Security Theorem)

That is, let us say we have two people, Person A and Person B, who are trying to communicate with one another. Specifically, Person A is observing samples, drawn one at a time, from some distribution X. Person A then wishes to communicate each sample to Person B. For example, XX might be a dice and Person A wishes to communicate to Person B the outcomes of repeated dice rolls.
This framework is depicted below:



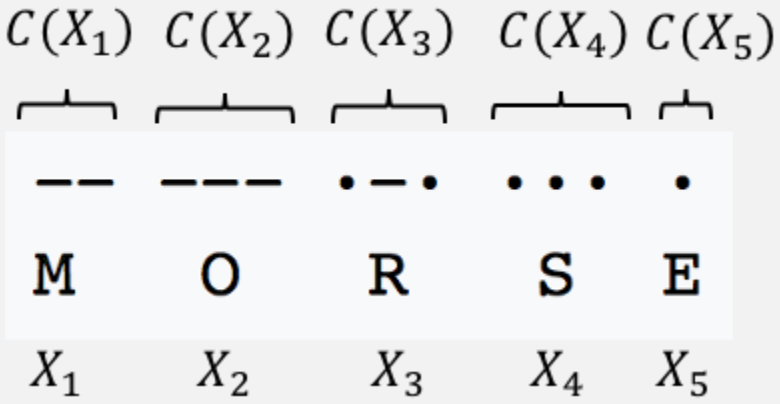
Let us rigorously describe the mathematical framework in which Person A is communicating messages to Person B. To do so, we will introduce a bunch of fundamental concepts from Coding Theory.
First, instead of calling each draw from X a "sample", we will instead call each draw a symbol. The idea here is that Person A is going to come up with some random sequence of symbols, each drawn from a distribution X, and attempt to communicate this random message, composed of the random sequence of symbols, to Person B. We'll call this random message the sequence of source symbols:

$$X_1, X_2, X_3, \ldots, X_m \overset{\text{i.i.d.}}{\sim} X$$

The idea of each Xi being a "symbol", intuitively implies that the number of outcomes that each Xi can take on is finite. To make this concrete, we will assert that X is a categorical distribution.

To make this concrete, let's look at a simplified, 2-ary version of Morse Code for encoding the English alphanumeric symbols into two symbols: "dots" and "dashes". In Morse Code, the source alphabet, X consists of the alphanumeric symbols, the code alphabet A consists of only a dot and dash, and the code function C maps each alphanumeric symbol to a sequence of dots and dashes. More specifically:

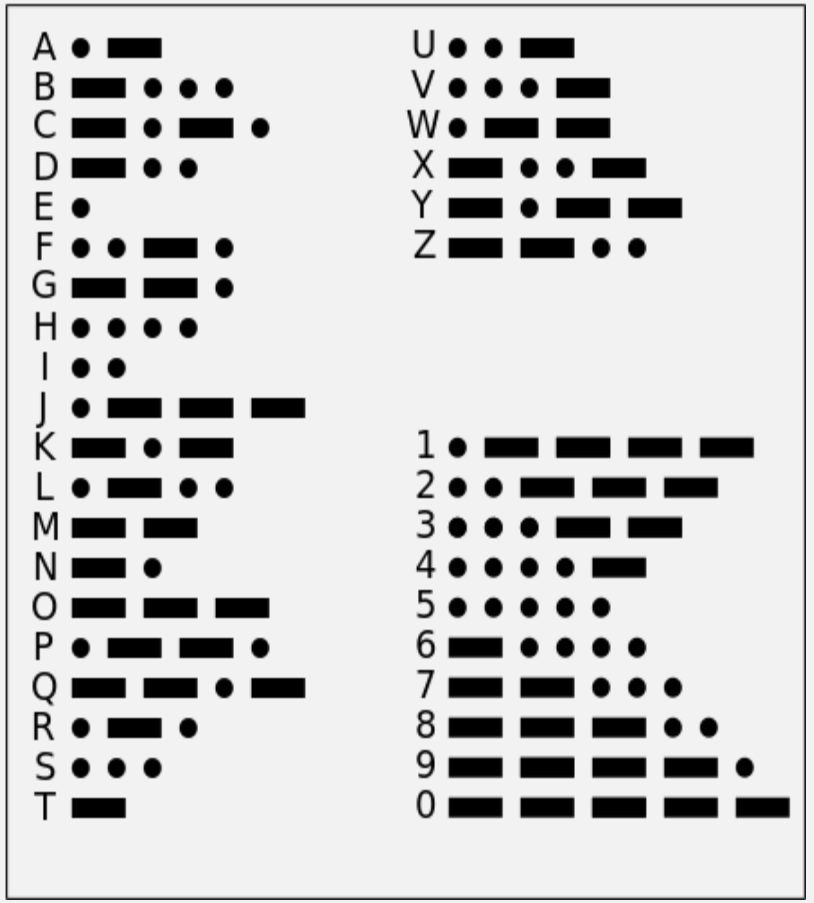
For example, the name "Morse" would be encoded as follows:



$$\mathcal{X} := \{A, B, \ldots, Z, 0,1, \ldots, 9\}$$

$$\mathcal{A} := \{\cdot, -\}$$

$C :=$



Shannon's Source Coding Theorem

Given some categorical distribution X, Shannon's Source Code Theorem tells us that no matter what C you choose, the smallest possible *expected code word length* is the entropy of X. That is,

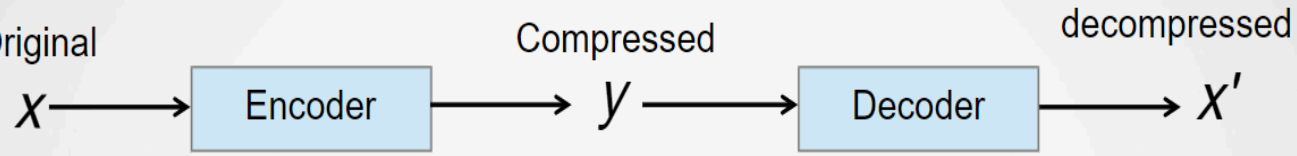
$$E\left[|C(X)|\right] = \sum_{x \in \mathcal{X}} |C(X)| P(X = x) \leq -\sum_{x \in \mathcal{X}} P(X = x) \log P(X = x) = H(X)$$

More formally:

Theorem 1 (Shannon's Source Coding Thoerem): Given a categorical random variable X over a finite source alphabet $\mathcal{X}$ and a code alphabet $\mathcal{A}$, then for all uniquely decodable $C : \mathcal{X} \to \mathcal{A}^*$, it holds that $E[|C(X)|] \geq H(X)$.

The expected code word length of X under some code C tells us how efficiently one can "compress" the information in X. If on average, C is able to produce small code words for each symbol drawn from X, then we are able to more efficiently communicate these symbols. Shannon's Source Coding Theorem tells us that the entropy of X is, in some sense, the true "information content" of the random variable because there is no C that will enable you to compress X past X's entropy.

The Noiseless Coding Theorem, Bit-rate (Data) Reduction, and Data Compression

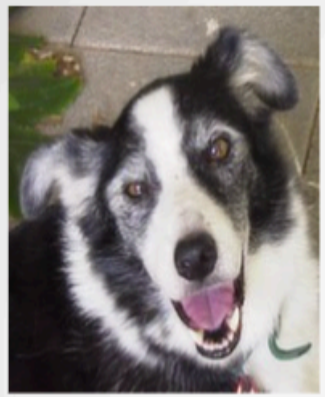


Lossless compression: $x = x'$

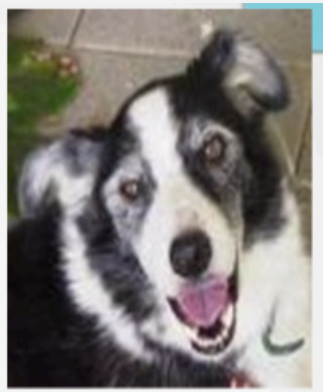
- (a.k.a. entropy coding, reversible coding)
  - Applications: text compression
    - "Do not send money" → "Do now send money"

Lossy compression: $x \neq x'$

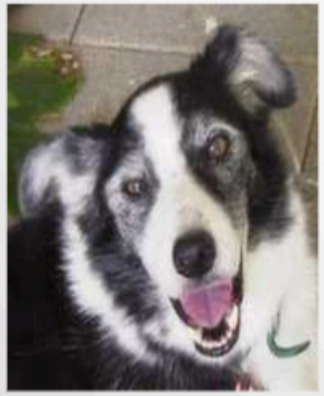
- (a.k.a. irreversible coding)
  - Applications: image compression
    - Keywords: distortion vs. perception



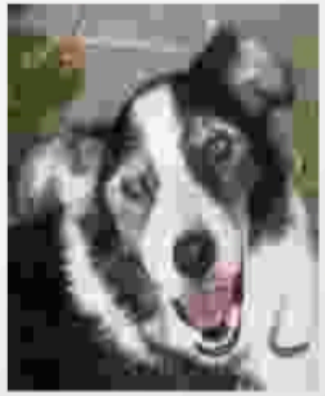
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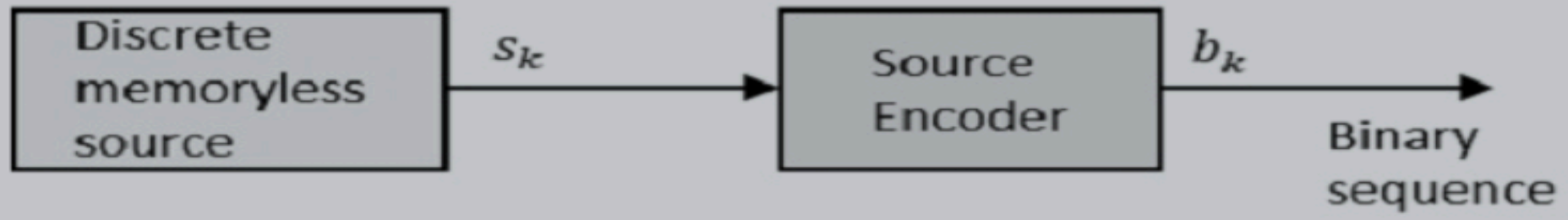


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The Code produced by a discrete memoryless source, has to be efficiently represented, which is an important problem in communications. For this to happen, there are code words, which represent these source codes.

For example, in telegraphy, we use Morse code, in which the alphabets are denoted by **Marks** and **Spaces**. If the letter **E** is considered, which is mostly used, it is denoted by **"."** Whereas the letter **Q** which is rarely used, is denoted by **"--.-"**

Let us take a look at the block diagram.



Where $\mathbf{S_k}$ is the output of the discrete memoryless source and $\mathbf{b_k}$ is the output of the source encoder which is represented by **0s** and **1s**.

The encoded sequence is such that it is conveniently decoded at the receiver.

Let us assume that the source has an alphabet with **k** different symbols and that the $\mathbf{k^{th}}$ symbol $\mathbf{S_k}$ occurs with the probability $\mathbf{P_k}$, where **k = 0, 1...k-1**.

Let the binary code word assigned to symbol $\mathbf{S_k}$, by the encoder having length $\mathbf{l_k}$, measured in bits.

Hence, we define the average code word length $\overline{L}$ of the source encoder as

$$\overline{L} = \sum_{k=0}^{k-1} p_k l_k$$

$\overline{L}$ represents the average number of bits per source symbol

$$\text{If} \quad L_{min} = \ minimum\, possible\, value\, of\, \overline{L}$$

Then **coding efficiency** can be defined as

$$\eta = \frac{Lmin}{\overline{L}}$$

$$\text{With} \quad \overline{L} \geq L_{min} \quad \text{we will have} \quad \eta \leq 1$$

However, the source encoder is considered efficient when $\eta = 1$

For this, the value L_{min} has to be determined.

Let us refer to the definition, *"Given a discrete memoryless source of entropy* $H(\delta)$ *, the average code-word length* $\overline{L}$ *for any source encoding is bounded as* $\overline{L} \geq H(\delta)$ *."*

In simpler words, the code word $example : Morsecodefortheword QUEUE is -.- --- ... -.$ is always greater than or equal to the source code $QUEUEinexample$. Which means, the symbols in the code word are greater than or equal to the alphabets in the source code.

Hence with $L_{min} = H(\delta)$, the efficiency of the source encoder in terms of Entropy $H(\delta)$ may be written as

$$\eta = \frac{H(\delta)}{\overline{L}}$$

This source coding theorem is called as **noiseless coding theorem** as it establishes an error-free encoding. It is also called as **Shannon's first theorem**.

The information theory (IT) frame work:

- ✔ Communication System
- ✔ Line Coding
- ✔ Channel Coding (Shannon's 2nd theorem)
- ✔ Cryptology (Shannon's security theory)
- ✔ Source Coding (Shannon's 1st theorem)

In information theory, Shannon's source coding theorem (or noiseless coding theorem) establishes the limits to possible data compression, and the operational meaning of the Shannon entropy.

THE MARKOV SOURCE

Each symbol depends on the previous one.

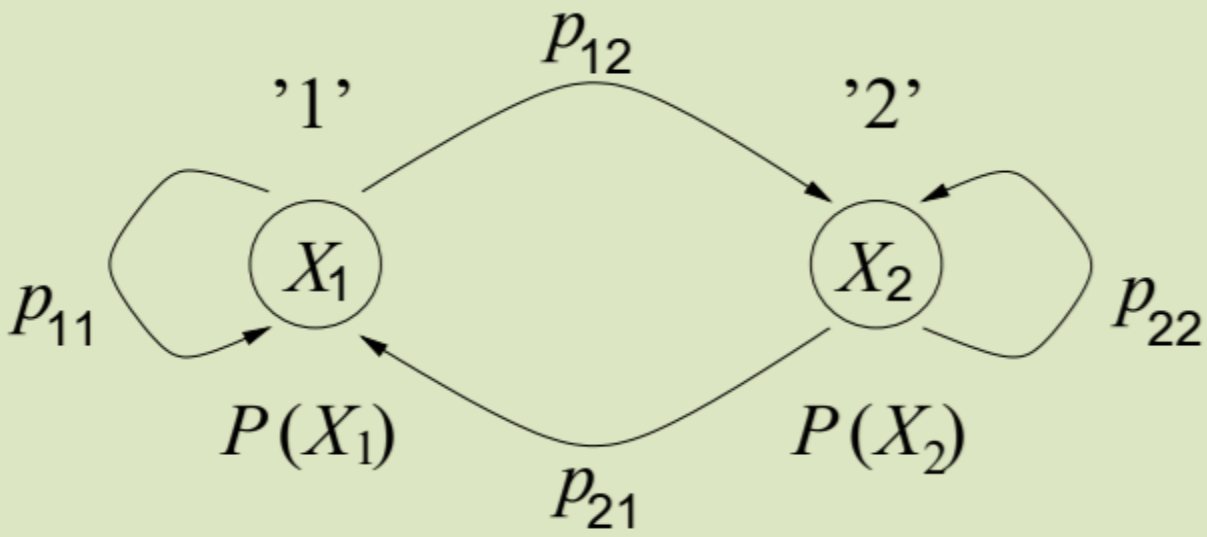
P(X1, X2, …, Xn) = P(X1) * P(X2|X1) * P(X3|X2) * … * P(Xn|Xn-1)

- Markov chains are used to connect various events or "states" with the "n" number of events that came previously.
- They work by having various states connected to other states that follow the current state.
- Each of these "transitions" has a weight that represents the likelihood of the second state following the first.

Here memory can be modelled by a **Markov process**

- Consider source with memory that emits a sequence of symbols $\{S(k)\}$ with "time" index k
- First order Markov process: the current symbol depends only on the previous symbol, $p(S(k)|S(k-1))$
- N-th order Markov process: the current symbol depends on N previous symbols, $p(S(k)|S(k-1), S(k-2), \cdots, S(k-N))$

- Source $S(k)$ can only generate two symbols, $X_1 = 1$ and $X_2 = 2$; their probability explicitly depends on the previous state (i.e. $p(S(k)|S(k-1))$)



- Probabilities of occurrence (prior probabilities) for states X_1 and X_2: $P_1 = P(X_1)$ and $P_2 = P(X_2)$ (i.e. $p(S(0) = 1) = P(X_1)$ and $p(S(0) = 2) = P(X_2)$)

- Transition probabilities: transition probabilities from state X_1 are given by the conditional probabilities $p_{12} = P(X_2|X_1)$ and $p_{11} = P(X_1|X_1) = 1 - P(X_2|X_1)$, etc. (i.e. $p(S(k) = j|S(k-1) = i) = p_{ij}$)

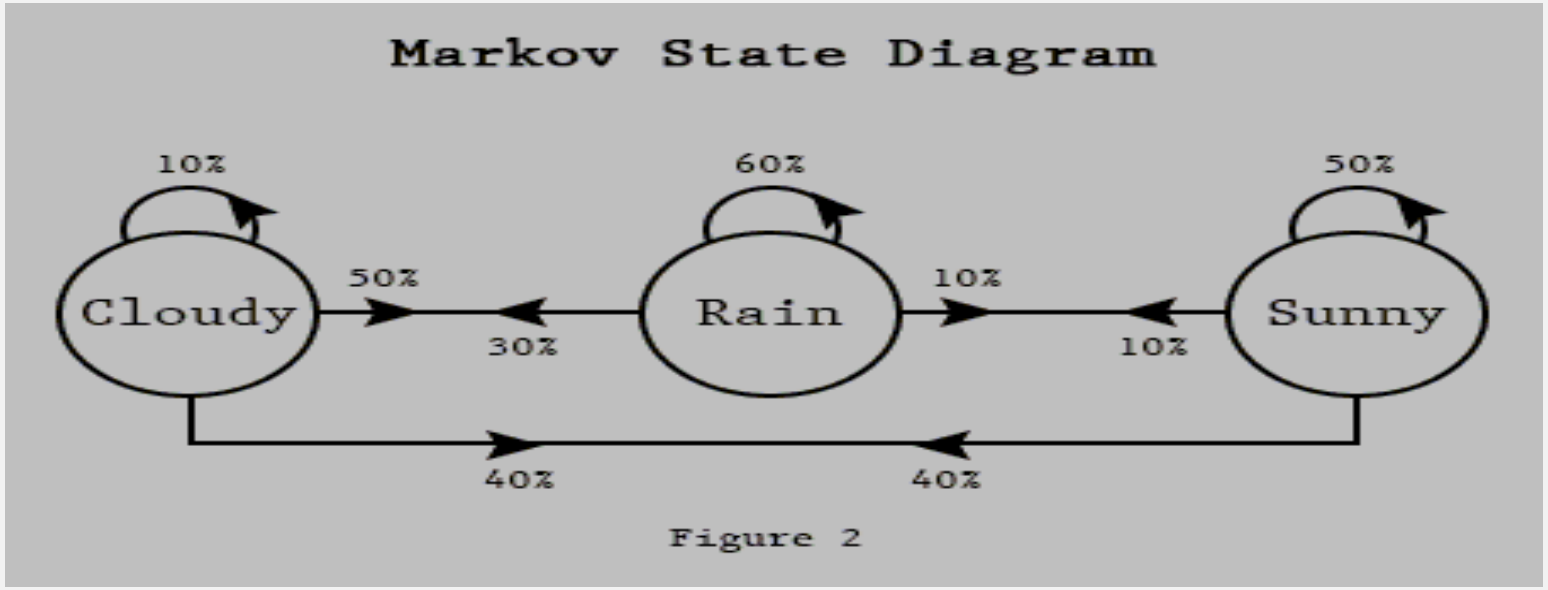
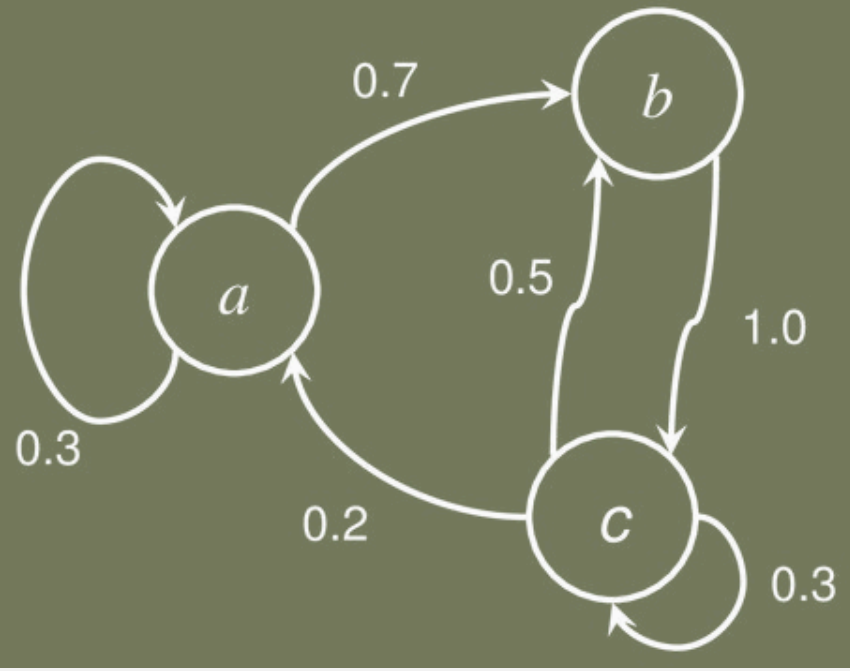


Figure 2

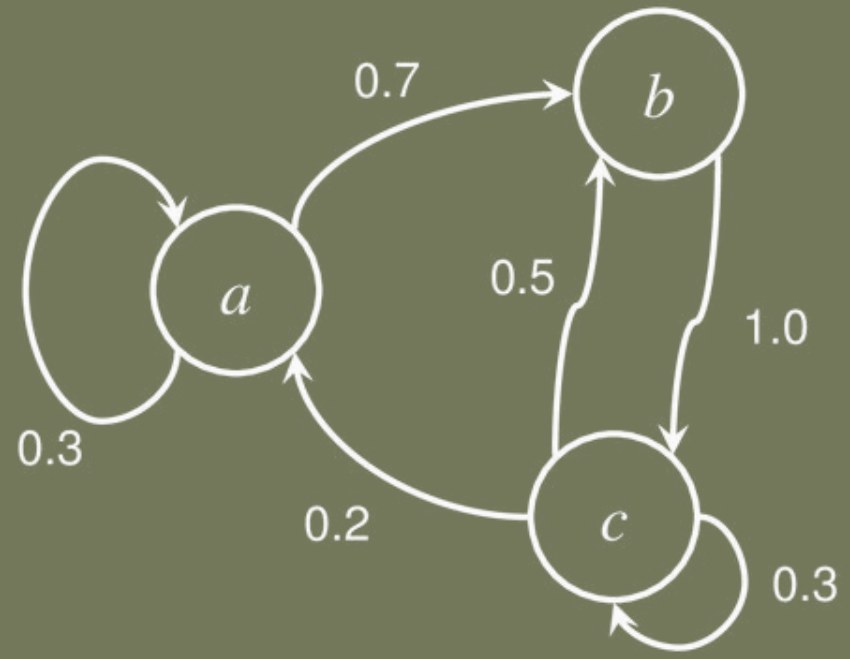
Using the diagram above, if the current day is cloudy, the percentage chance it will be cloudy the following day is 10%. Following this diagram for the rest of the weather outcomes, the percentage chance it will be rainy the following day is 50%, and the percentage chance it will be sunny the following day is 40%. This information is solely based on past information gathered from historical data. However, lots of other things can influence these percentage chances such as natural disasters or out of the ordinary weather events. Markov chains have been commonly used in weather models, but are also used in other areas like financial modeling and population dynamics.

- A symbol depends only on the previous symbol, so the source can be modelled by a state diagram.



A ternary source with alphabet $\mathcal{A} = (a, b, c)$.

- Assume we are in state a, i.e., $X_k = a$.
- The probabilities for the next symbol are:

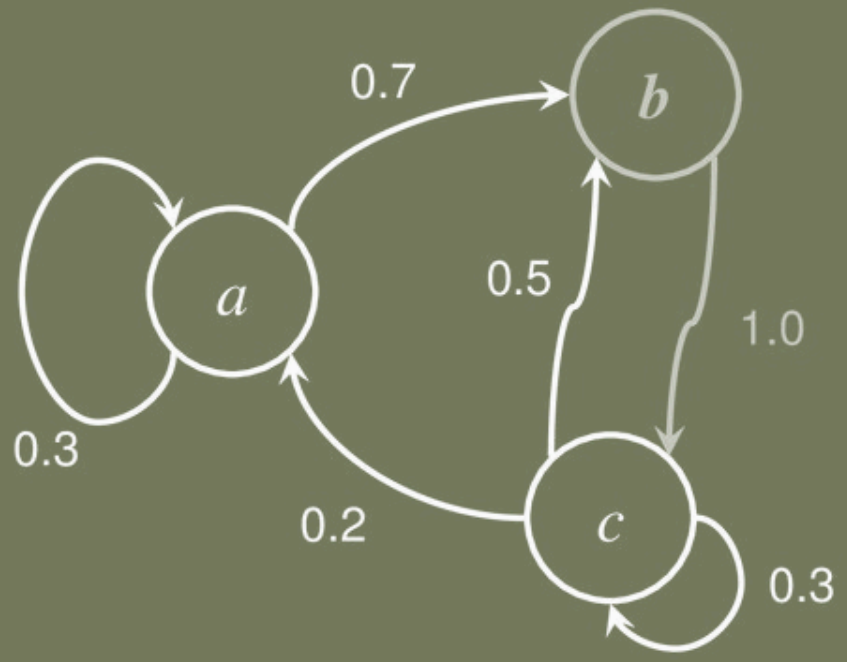


$$P(X_{k+1} = a \mid X_k = a) = 0.3$$

$$P(X_{k+1} = b \mid X_k = a) = 0.7$$

$$P(X_{k+1} = c \mid X_k = a) = 0$$

- So, if $X_{k+1} = b$, we know that X_{k+2} will equal c.



$$P(X_{k+2} = a \mid X_{k+1} = b) = 0$$

$$P(X_{k+2} = b \mid X_{k+1} = b) = 0$$

$$P(X_{k+2} = c \mid X_{k+1} = b) = 1$$

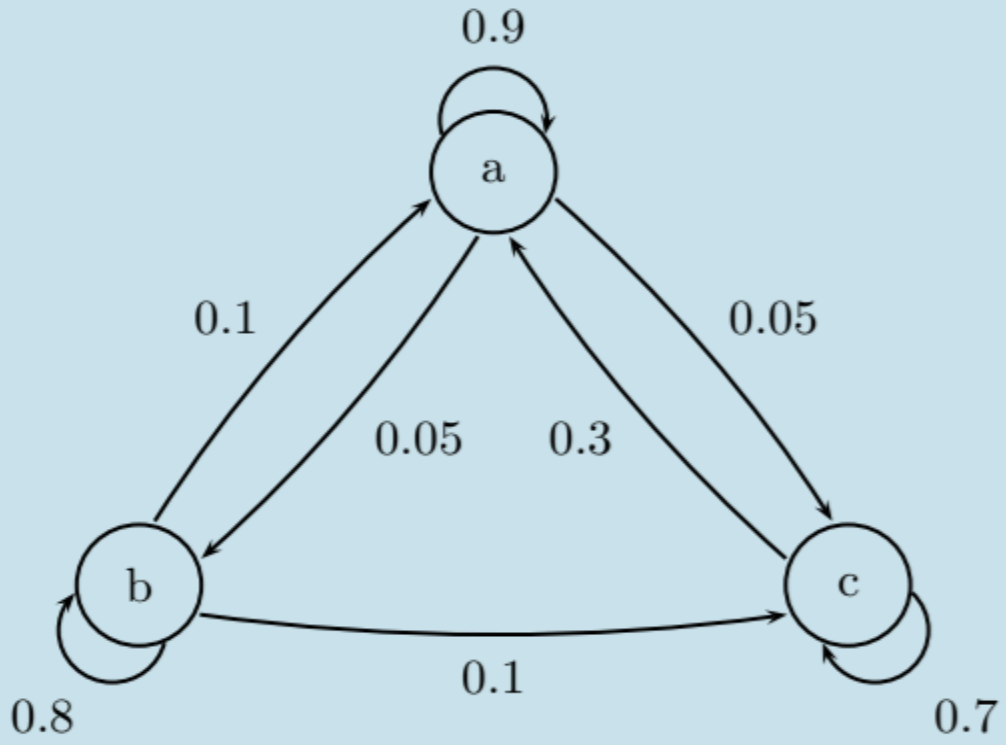
Suppose we have a stationary Markov source X_n of order 1 with the alphabet $\mathcal{A} = \{a, b, c\}$ and the transition probabilities $P(x_n|x_{n-1})$ given by

$$\begin{array}{lll} P(a|a) = 0.9 & P(b|a) = 0.05 & P(c|a) = 0.05 \\ P(a|b) = 0.1 & P(b|b) = 0.8 & P(c|b) = 0.1 \\ P(a|c) = 0.3 & P(b|c) = 0 & P(c|c) = 0.7 \end{array}$$

Suppose we have a stationary Markov source X_n of order 1 with the alphabet $\mathcal{A} = \{a, b, c\}$ and the transition probabilities $P(x_n|x_{n-1})$ given by

$$\begin{array}{lll} P(a|a) = 0.9 & P(b|a) = 0.05 & P(c|a) = 0.05 \\ P(a|b) = 0.1 & P(b|b) = 0.8 & P(c|b) = 0.1 \\ P(a|c) = 0.3 & P(b|c) = 0 & P(c|c) = 0.7 \end{array}$$

We can draw the Markov source as the state graph below:



Problem 1

Consider the Markov chain with three states, $S = \{1, 2, 3\}$, that has the following transition matrix

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}.$$

a. Draw the state transition diagram for this chain.

b. If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

a. The state transition diagram is shown in Figure 11.6

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}.$$

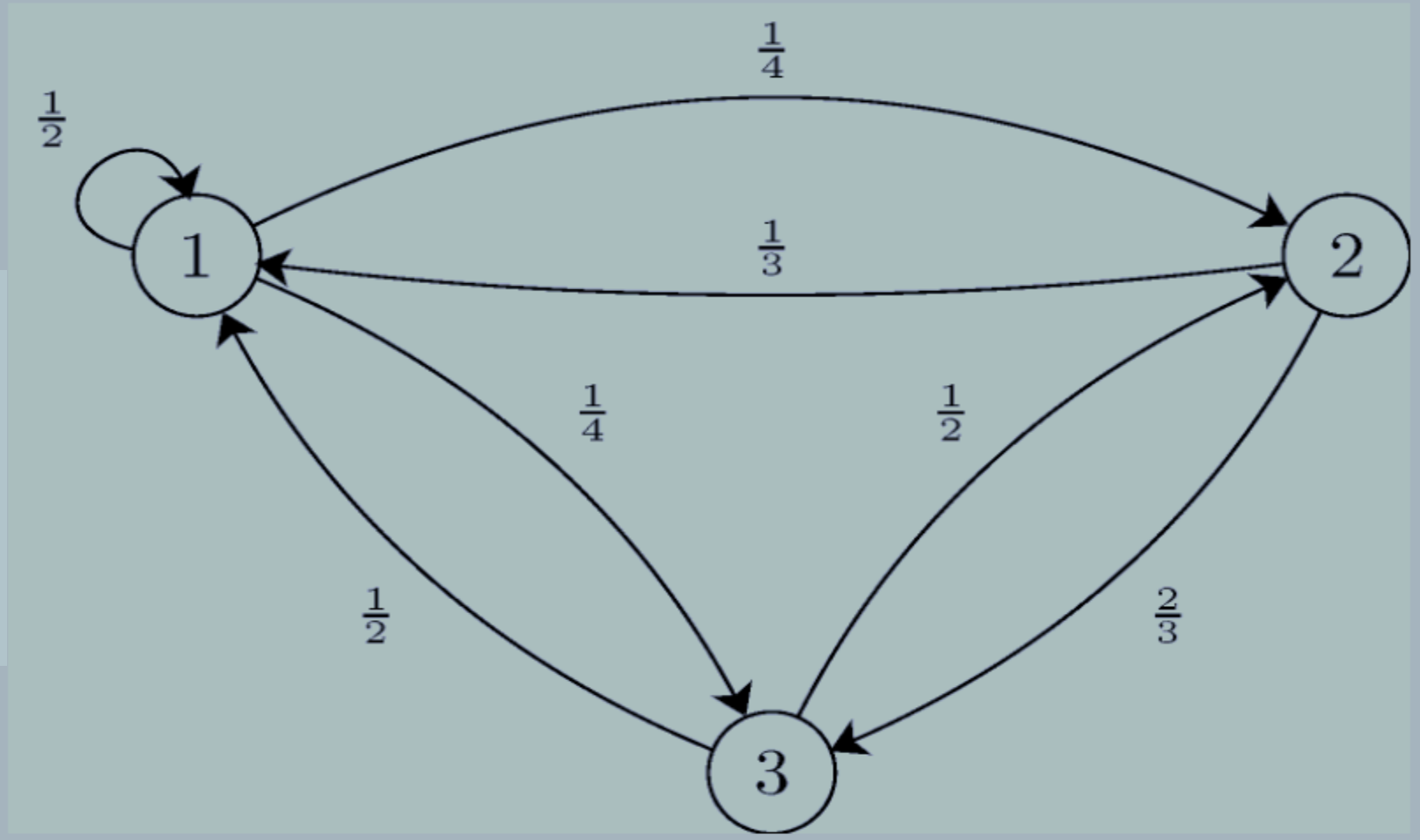


Figure 11.6 - A state transition diagram.

b. If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$, find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.

b. First, we obtain

$$\begin{aligned} P(X_1 = 3) &= 1 - P(X_1 = 1) - P(X_1 = 2) \\ &= 1 - \frac{1}{4} - \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

We can now write

$$\begin{aligned} P(X_1 = 3, X_2 = 2, X_3 = 1) &= P(X_1 = 3) \cdot p_{32} \cdot p_{21} \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} \\ &= \frac{1}{12}. \end{aligned}$$

Problem 2

Consider the Markov chain in Figure 11.17. There are two recurrent classes, $R_1 = \{1, 2\}$, and $R_2 = \{5, 6, 7\}$. Assuming $X_0 = 3$, find the probability that the chain gets absorbed in R_1.

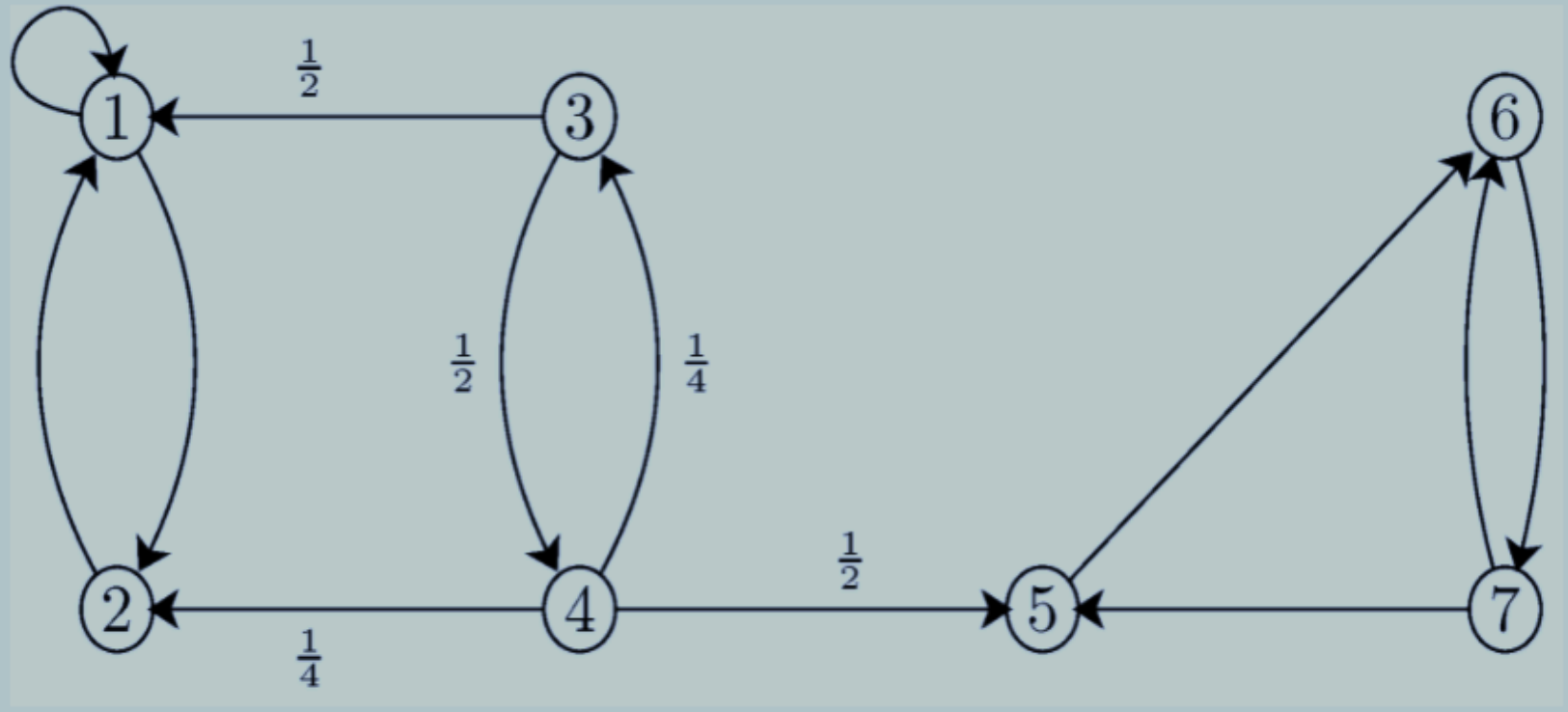


Figure 11.17 - A state transition diagram.

Here, we can replace each recurrent class with one absorbing state. The resulting state diagram is shown in Figure 11.18

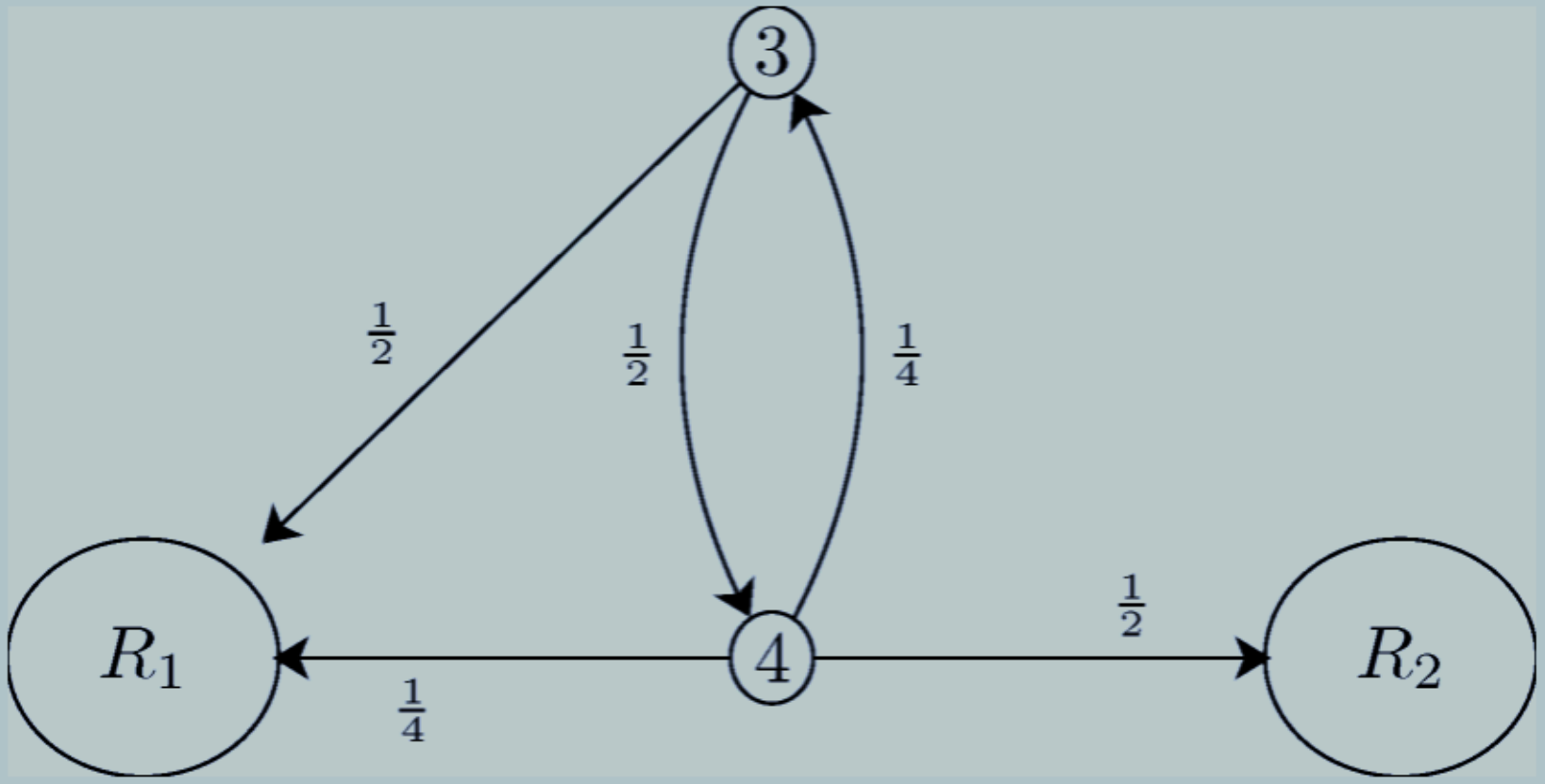


Figure 11.18 - The state transition diagram in which we have replaced each recurrent class with one absorbing state.

Now we can apply our standard methodology to find probability of absorption in state R_1. In particular, define

$$a_i = P(\text{absorption in } R_1 | X_0 = i), \quad \text{for all } i \in S.$$

By the above definition, we have $a_{R_1} = 1$, and $a_{R_2} = 0$. To find the unknown values of a_i's, we can use the following equations

$$a_i = \sum_k a_k p_{ik}, \quad \text{for } i \in S.$$

We obtain

$$\begin{aligned} a_3 &= \frac{1}{2} a_{R_1} + \frac{1}{2} a_4 \\ &= \frac{1}{2} + \frac{1}{2} a_4, \end{aligned}$$

$$\begin{aligned} a_4 &= \frac{1}{4} a_{R_1} + \frac{1}{4} a_3 + \frac{1}{2} a_{R_2} \\ &= \frac{1}{4} + \frac{1}{4} a_3. \end{aligned}$$

Solving the above equations, we obtain

$$a_3 = \frac{5}{7}, \quad a_4 = \frac{3}{7}.$$

Therefore, if $X_0 = 3$, the chain will end up in class R_1 with probability $a_3 = \frac{5}{7}$.

LECTURE – 5

Joint Entropy

Given two discrete memoryless sources X and Y with alphabet $\mathcal{X}$ and $\mathcal{Y}$ respectively, we define the information obtained by observing the couple of random variables (X, Y). The extension of the entropy definition to a pair of random variables is called *joint entropy* and involves the joint distribution $p_{XY}(x, y)$, that is the statistical quantity describing the dependence between the variables.

Definition. The joint entropy $H(X, Y)$ of a pair of discrete random variables (X, Y) with joint distribution $p(x, y)$ is defined as

$$H(X, Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y). \tag{2.1}$$

The joint entropy represents the amount of information needed on average to specify the value of two discrete random variables.

Conditional Entropy

We now characterize the information received by observing a random variable X when we already know the value taken by another random variable Y. Reasonably, if the knowledge of Y gives us some information about X, the information carried by X will no more be $H(X)$.
Given a single realization $Y = y$, we define the entropy of the conditional distribution $p(x|y)$[1] as

$$H(X|Y = y) = -\sum_{x \in \mathcal{X}} p(x|y) \log p(x|y). \tag{2.6}$$

Definition. Given a pair of random variables (X, Y), the conditional entropy $H(X|Y)$ is defined as

$$H(X|Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x|y). \tag{2.7}$$

The conditional entropy can be expressed as the expected value of the entropies of the conditional distributions averaged over the conditioning random variable Y, i.e. $H(X|Y) = \sum_{y \in \mathcal{Y}} p(y) \cdot H(X|Y = y)$.

- Using the input probabilities P (x_i), output probabilities P (y_j), transition probabilities P (y_j/x_i) and joint probabilities P (x_i, y_j), various entropy functions for a channel with m inputs and n outputs are defined

$$H(X) = -\sum_{i=1}^{m} P(x_i)\log_2 p(x_i)$$

$$H(Y) = -\sum_{j=1}^{n} P(y_j)\log_2 p(y_j)$$

$$H(X|Y) = -\sum_{j=1}^{n}\sum_{i=1}^{m} P(x_i, y_j)\log_2 P(x_i|y_j)$$

$$H(Y|X) = -\sum_{j=1}^{n}\sum_{i=1}^{m} P(x_i, y_j)\log_2 P(y_i|x_j)$$

$$H(X, |Y) = -\sum_{j=1}^{n}\sum_{i=1}^{m} P(x_i, y_j)\log_2 P(x_i, y_j)$$

- H (X) is the average uncertainty of the channel input and H (Y) is the average uncertainty of the channel output.
- The conditional entropy H (X/Y) is a measure of the average uncertainty remaining about the channel input after the channel output has been observed. H (X/Y) is also called equivocation of X w.r.t. Y.
- The conditional entropy H (Y/X) is the average uncertainty of the channel output given that X was transmitted.

- The joint entropy H (X, Y) is the average uncertainty of the communication channel as a whole. Few useful relationships among the above various entropies are as under:

a. $H(X, Y) = H(X/Y) + H(Y)$

b. $H(X, Y) = H(Y/X) + H(X)$

c. $H(X, Y) = H(X) + H(Y)$

d. $H(X/Y) = H(X, Y) - H(Y)$

- X and Y are statistically independent
- The conditional entropy or conditional uncertainty of X given random variable Y is the average conditional entropy over Y.
- The joint entropy of two discrete random variables X and Y is merely the entropy of their pairing: (X, Y), this implies that if X and Y are independent, then their joint entropy is the sum of their individual entropies.

The Mutual Information:

- Mutual information measures the amount of information that can be obtained about
  one random variable by observing another.
- It is important in communication where it can be used to maximize the amount of information shared between sent and received signals.
- The mutual information denoted by I (X, Y) of a channel is defined by:

$$I(X;Y) = H(X) - H(X|Y) \text{ bits/symbol}$$

- Since H (X) represents the uncertainty about the channel input before the channel output is observed and H (X/Y) represents the uncertainty about the channel input after the channel output is observed, the mutual information I (X; Y) represents the uncertainty about the channel input that is resolved by observing the channel output.

Problem 1

Now assume that you have the joint probability of a vowel and a consonant occurring together in the same syllable:

$f(x,y)$	p	t	k	$f(y)$
a	$\frac{1}{16}$	$\frac{3}{8}$	$\frac{1}{16}$	$\frac{1}{2}$
i	$\frac{1}{16}$	$\frac{3}{16}$	0	$\frac{1}{4}$
u	0	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
$f(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$	

Compute the conditional probabilities; for example:

$$f(a|p) = \frac{f(a,p)}{f(p)} = \frac{\frac{1}{16}}{\frac{1}{8}} = \frac{1}{2}$$

$$f(a|t) = \frac{f(a,t)}{f(t)} = \frac{\frac{3}{8}}{\frac{3}{4}} = \frac{1}{2}$$

$f(x,y)$	p	t	k	$f(y)$
a	$\frac{1}{16}$	$\frac{3}{8}$	$\frac{1}{16}$	$\frac{1}{2}$
i	$\frac{1}{16}$	$\frac{3}{16}$	0	$\frac{1}{4}$
u	0	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
$f(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$	

Now compute the conditional entropy of a vowel given a consonant:

$$
\begin{aligned}
H(V|C) &= -\sum_{x\in C}\sum_{y\in V} f(x,y)\log f(y|x) \\
&= -(f(a,p)\log f(a|p) + f(a,t)\log f(a|t) + f(a,k)\log f(a|k) + \\
&\quad f(i,p)\log f(i|p) + f(i,t)\log f(i|t) + f(i,k)\log f(i|k) + \\
&\quad f(u,p)\log f(u|p) + f(u,t)\log f(u|t) + f(u,k)\log f(u|k)) \\
&= -(\tfrac{1}{16}\log\tfrac{\frac{1}{16}}{\frac{1}{8}} + \tfrac{3}{8}\log\tfrac{\frac{3}{8}}{\frac{3}{4}} + \tfrac{1}{16}\log\tfrac{\frac{1}{16}}{\frac{1}{8}} + \\
&\quad \tfrac{1}{16}\log\tfrac{\frac{1}{16}}{\frac{1}{8}} + \tfrac{3}{16}\log\tfrac{\frac{3}{16}}{\frac{3}{4}} + 0 + \\
&\quad 0 + \tfrac{3}{16}\log\tfrac{\frac{3}{16}}{\frac{3}{4}} + \tfrac{1}{16}\log\tfrac{\frac{1}{16}}{\frac{1}{8}}) \\
&= \tfrac{11}{8} = 1.375 \text{ bits}
\end{aligned}
$$

For probability distributions we defined:

$$f(y|x) = \frac{f(x,y)}{g(x)}$$

A similar theorem holds for entropy:

Theorem: Conditional Entropy

If X and Y are discrete random variables with joint entropy $H(X,Y)$ and the marginal entropy of X is $H(X)$, then:

$$H(Y|X) = H(X,Y) - H(X)$$

Division instead of subtraction as entropy is defined on logarithms.

$f(x,y)$	p	t	k	$f(y)$
a	$\frac{1}{16}$	$\frac{3}{8}$	$\frac{1}{16}$	$\frac{1}{2}$
i	$\frac{1}{16}$	$\frac{3}{16}$	0	$\frac{1}{4}$
u	0	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
$f(x)$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{8}$	

Use the previous theorem to compute the joint entropy of a consonant and a vowel. First compute $H(C)$:

$$\begin{aligned} H(C) &= -\sum_{x \in C} f(x) \log f(x) \\ &= -(f(p) \log f(p) + f(t) \log f(t) + f(k) \log f(k)) \\ &= -(\frac{1}{8} \log \frac{1}{8} + \frac{3}{4} \log \frac{3}{4} + \frac{1}{8} \log \frac{1}{8}) \\ &= 1.061 \text{ bits} \end{aligned}$$

Then we can compute the joint entropy as:

$$H(V, C) = H(V|C) + H(C) = 1.375 + 1.061 = 2.436 \text{ bits}$$

- Entropy measures the amount of information in a random variable or the length of the message required to transmit the outcome;
- joint entropy is the amount of information in two (or more) random variables;
- conditional entropy is the amount of information in one random variable given we already know the other.

Source Coding:

- **Definition:** A conversion of the output of a discrete memory less source (DMS) into a sequence of binary symbols i.e. binary code word, is called **Source Coding**.
- The device that performs this conversion is called the **Source Encoder**.
- **Objective of Source Coding:** An objective of source coding is to minimize the average bit rate required for representation of the source by reducing the redundancy of the information source

The Source Coding Theorem

- The source coding theorem states that for a DMS X, with entropy H (X), the average code word length $\bar{L}$ per symbol is bounded as $\bar{L} \geq H(X)$
- And further, $\bar{L}$ can be made as close to H (X) as desired for some suitable chosen code
- Thus,

$$\bar{L}_{min} = H(X)$$

- The code efficiency can be rewritten as

$$\eta = \frac{H(X)}{\bar{L}}$$

$H \leq L < H+1$

Classification of Code

1. Fixed – Length Codes
2. Variable – Length Codes
3. Distinct Codes
4. Prefix – Free Codes
5. Uniquely Decodable Codes
6. Instantaneous Codes
7. Optimal Codes

x_i	Code 1	Code 2	Code 3	Code 4	Code 5	Code 6
x_1	00	00	0	0	0	1
x_2	01	01	1	10	01	01
x_3	00	10	00	110	011	001
x_4	11	11	11	111	0111	0001

1. **Fixed – Length Codes:** A fixed – length code is one whose code word length is fixed. Code 1 and Code 2 of above table are fixed – length code words with length 2.
2. **Variable – Length Codes:** A variable – length code is one whose code word length is not fixed. All codes of above table except Code 1 and Code 2 are variable – length codes.
3. **Distinct Codes:** A code is distinct if each code word is distinguishable from each other. All codes of above table except Code 1 are distinct codes.
4. **Prefix – Free Codes:** A code in which no code word can be formed by adding code symbols to another code word is called a prefix-free code. In a prefix – free code, no code word is prefix of another. Codes 2, 4 and 6 of above table are prefix – free codes.
5. **Uniquely Decodable Codes:** A distinct code is uniquely decodable if the original source sequence can be reconstructed perfectly from the encoded binary sequence. A sufficient condition to ensure that a code is uniquely decodable is that no code word is a prefix of another. Thus the prefix – free codes 2, 4 and 6 are uniquely decodable codes. Prefix – free condition is not a necessary condition for uniquely decidability. Code 5 albeit does not satisfy the prefix – free condition and yet it is a uniquely decodable code since the bit 0 indicates the beginning of each code word of the code.
6. **Instantaneous Codes:** A uniquely decodable code is called an instantaneous code if the end of any code word is recognizable without examining subsequent code symbols. The instantaneous codes have the property previously mentioned that no code word is a prefix of another code word. Prefix – free codes are sometimes known as instantaneous codes.
7. **Optimal Codes:** A code is said to be optimal if it is instantaneous and has the minimum average L for a given source with a given probability assignment for the source symbols

Example : A scanner converts a black and white document, line-by-line into binary data for transmission. The scanner produces source data comprising symbols representing runs of up to six similar image pixel elements with the probabilities as shown below:

-

No. of consecutive pixels	1	2	3	4	5	6
Probability of occurrence	0.2	0.4	0.15	0.1	0.06	0.09

Determine the average length of a run (in pixels) and the corresponding effective information rate for this source when the scanner is traversing 1000 pixel/s.

$$H = \sum_{m=1}^{6} P(m) \log_2 \frac{1}{P(m)}$$

$$= 0.2 \times 2.32 + 0.4 \times 1.32 + 0.15 \times 2.74 + 0.1 \times 3.32 + 0.06 \times 4.06 + 0.09 \times 3.47$$

$$= 2.29 \quad \text{bit/symbol}$$

Average length is

$$L = \sum_{m=1}^{6} P(m)\, l_m = 0.2 + 0.8 + 0.45 + 0.4 + 0.3 + 0.54 = 2.69 \quad \text{pixels}$$

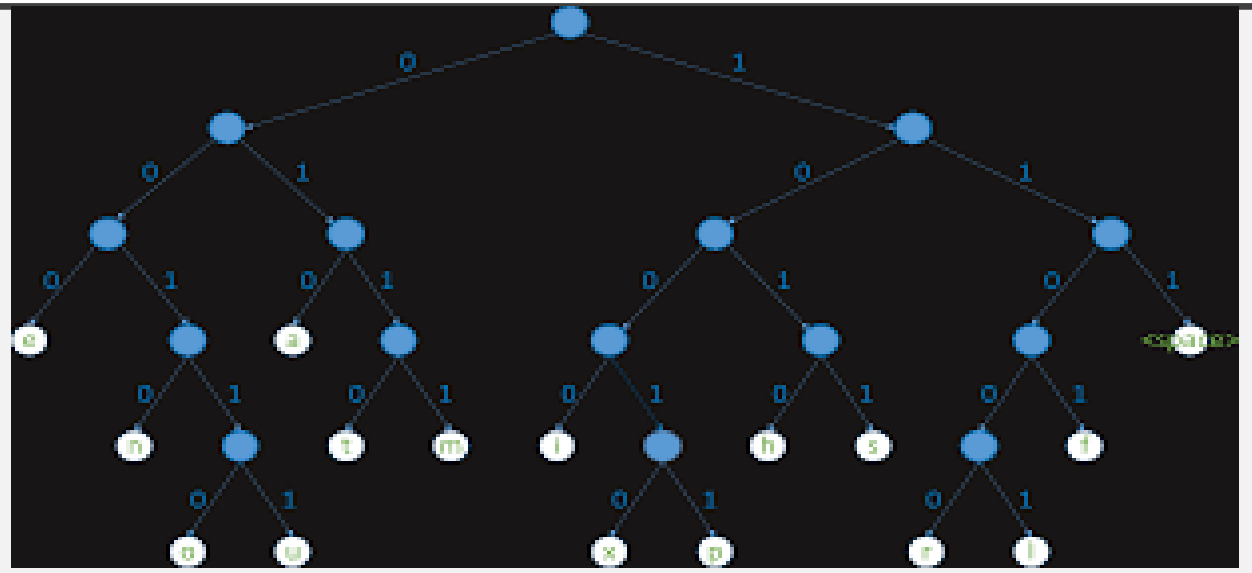
At 1000 pixel/s scan rate we generate 1000/2.69 = 372 symbol/s. Thus the source information rate is 2.29 x 372 = 852 bit/s.

We are generally interested in finding a more efficient code which represents the same information using fewer digits on average. This results in different lengths of codeword being used for different symbols. The problem with such variable length codes is in recognizing the start and end of the symbols.

$$H = \sum_{m=1}^{M} P(m) \log_2 \frac{1}{P(m)} \quad \text{(bit/symbol)}$$

$$L = \sum_{m=1}^{M} P(m)\, l_m \quad \text{(binary digits/symbol)}$$

LECTURE – 6



Huffman Code

- Huffman's Coding algorithms is used for compression of data so that it doesn't lose any information.
- Each symbol is converted into a binary code.
- In order to decompress the data and see the initial symbols, we need the frequencies of elements and the compressed data.
- Huffman Coding uses prefix rules which assures that there is no ambiguity in the decoding process.
- The prefix rule states that no code is a prefix of another code.
- **This algorithm includes two parts: Building the Huffman Tree from the input characters; and Traversing the tree to assign codes to symbols.**
- With the Huffman code in the binary case the two least probable source output symbols are joined together, resulting in a new message alphabet with one less symbol.

1. **take together smallest probabilities: P(i) + P(j)**
2. **replace symbol i and j by new symbol**
3. **go to 1 - until end**

- Application examples: JPEG, MPEG, MP3

ADVANTAGES:

- ***uniquely*** decodable code
- ***smallest*** average codeword length

DISADVANTAGES:

- LARGE tables give complexity
- sensitive to channel errors

Huffman coding results in an optimal code. It is the code that has the highest efficiency.

The Huffman coding procedure is as follows:

I. List the source symbols in order of decreasing probability.

I. Combine the probabilities of the two symbols having the lowest probabilities and reorder the resultant probabilities, this step is called reduction 1. The same procedure is repeated until there are two ordered probabilities remaining.

I. Start encoding with the last reduction, which consists of exactly two ordered probabilities. Assign 0 as the first digit in the code word for all the source symbols associated with the first probability; assign 1 to the second probability.

I. Now go back and assign 0 and 1 to the second digit for the two probabilities that were combined in the previous reduction step, retaining all the source symbols associated with the first probability; assign 1 to the second probability.

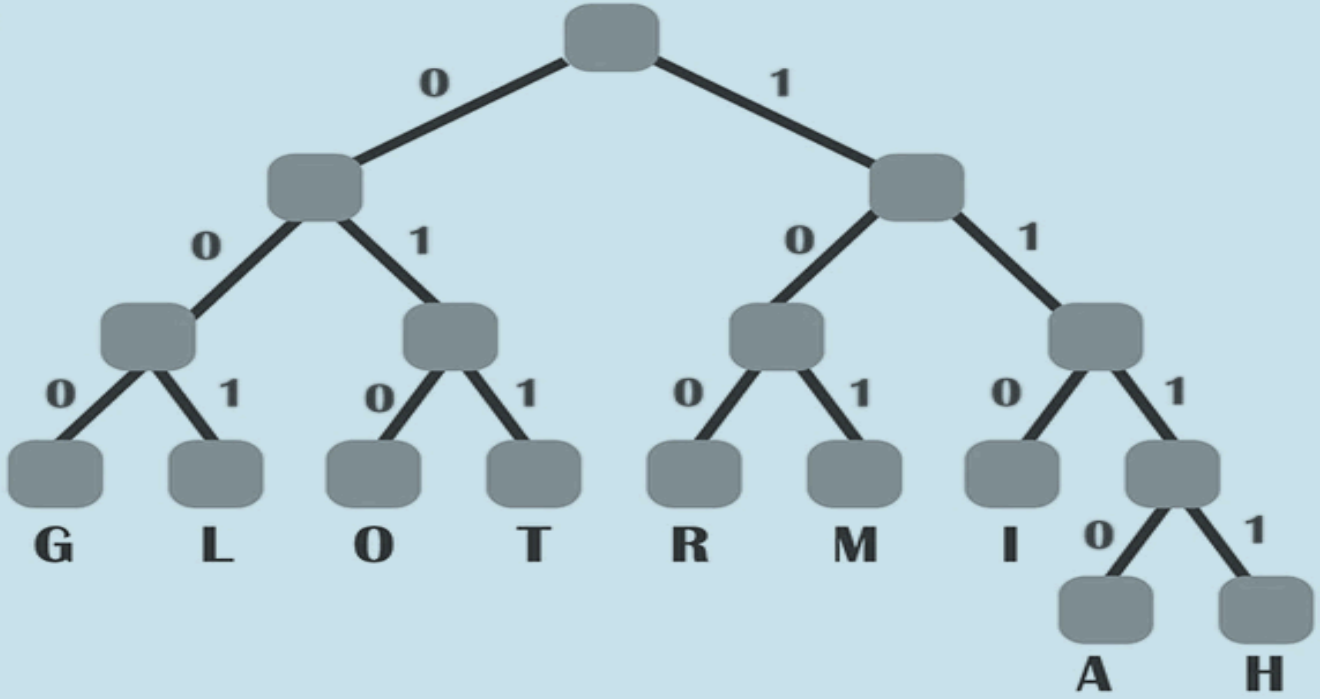
I. Keep regressing this way until the first column is reached.

I. The code word is obtained tracing back from right to left.



Huffman Coding

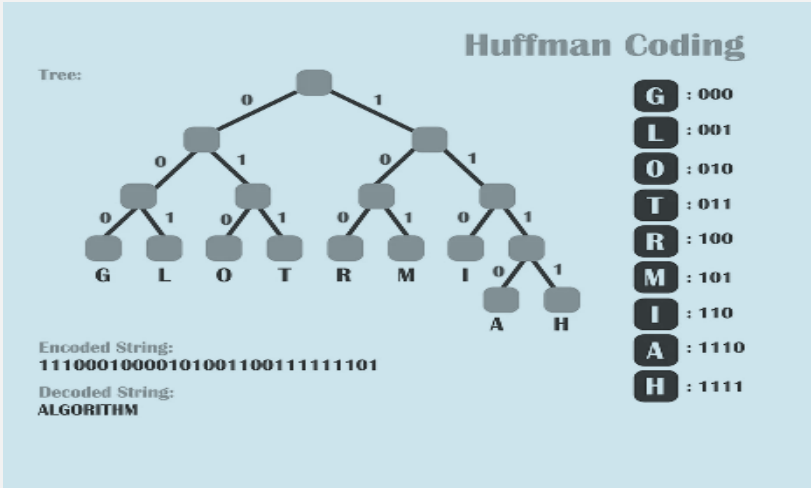
Tree:



G : 000
L : 001
O : 010
T : 011
R : 100
M : 101
I : 110
A : 1110
H : 1111

Encoded String:
11100010000101001100111111101

Decoded String:
ALGORITHM


Huffman Coding
Tree:
G L O T R M I A H
G : 000
L : 001
O : 010
T : 011
R : 100
M : 101
I : 110
A : 1110
H : 1111
Encoded String:
11100010000101001100111111101
Decoded String:
ALGORITHM

Problem Statement

In a communication system, the source transmits five different messages say x1, x2, x3, x4, x5 with the probabilities of 0.3, 0.15, 0.25, 0.05, 0.25 respectively. If the number of symbols used to code these messages is 2, then find the code word for each message and the coding efficiency by applying Huffman Coding method.

In a communication system, the source transmits five different messages say x1, x2, x3, x4, x5 with the probabilities of 0.3, 0.15, 0.25, 0.05, 0.25 respectively. If the number of symbols used to code these messages is 2, then find the code word for each message and the coding efficiency by applying Huffman Coding method.

Given:

[X] = [x1 x2 x3 x4 x5]

[P] = [0.3 0.15 0.25 0.05 0.25]

M = 2.

1) Arrange given messages in decreasing order of probabilities

[X] = [x1 x2 x3 x4 x5]

[P] = [0.3 0.15 0.25 0.05 0.25]

Message	Probability
x1	0.3
x3	0.25
x5	0.25
x2	0.15
x4	0.05

2) Group the least probable M messages and assign them symbols used for coding.

Here, M=2, Let symbols assigned be 0 and 1

Message	Probability	
x1	0.3	
x3	0.25	
x5	0.25	
x2	0.15	**0**
x4	0.05	**1**

3) Add the probabilities of grouped messages and place them as high as possible and rearrange them in decreasing order again. This process is known as reduction.

Message	Probability	1st Reduction	
x1	0.3	0.3	
x3	0.25	0.25	
x5	0.25	0.25	
x2	0.15	**0**	→ 0.2
x4	0.05	**1**	

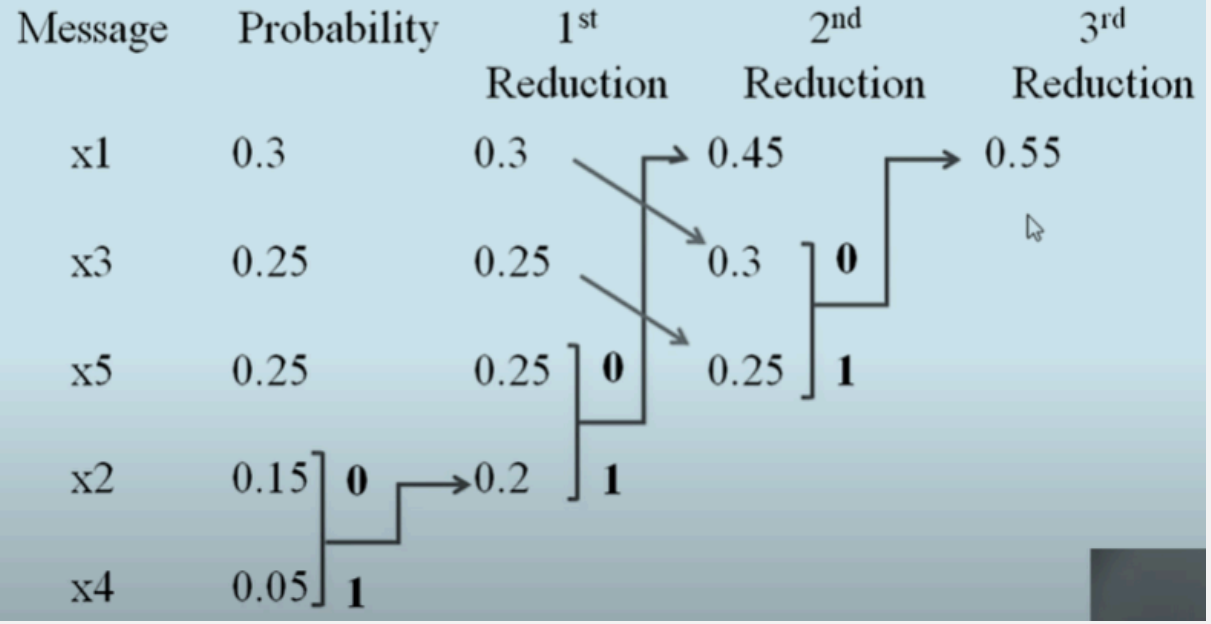
4) Repeat the steps 2 and 3 till only M probabilities remain

Message	Probability	1st Reduction	2nd Reduction
x1	0.3	0.3	0.45
x3	0.25	0.25	
x5	0.25	0.25 **0**	
x2	0.15 **0**	0.2 **1**	
x4	0.05 **1**		

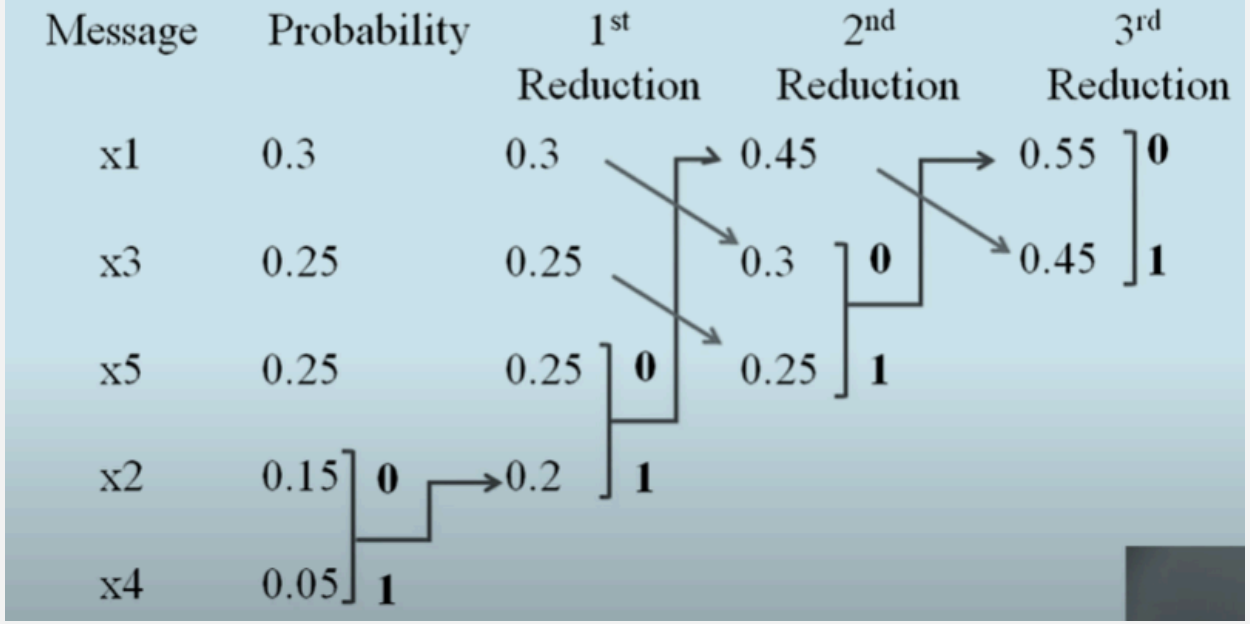
4) Repeat the steps 2 and 3 till only M probabilities remain

Message	Probability	1st Reduction	2nd Reduction
x1	0.3	0.3	0.45
x3	0.25	0.25	0.3
x5	0.25	0.25 **0**	0.25
x2	0.15 **0**	0.2 **1**	
x4	0.05 **1**		

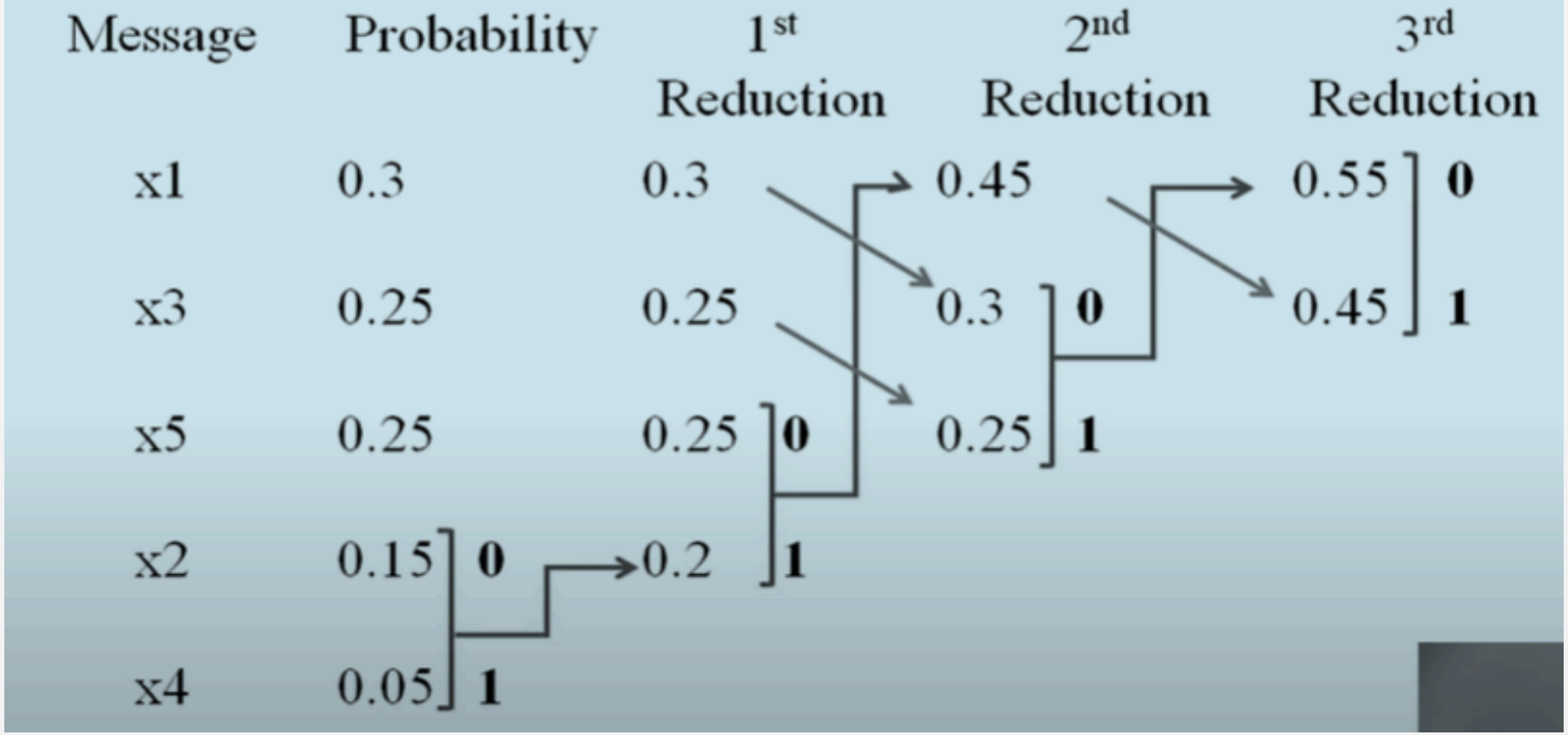
4) Repeat the steps 2 and 3 till only M probabilities remain

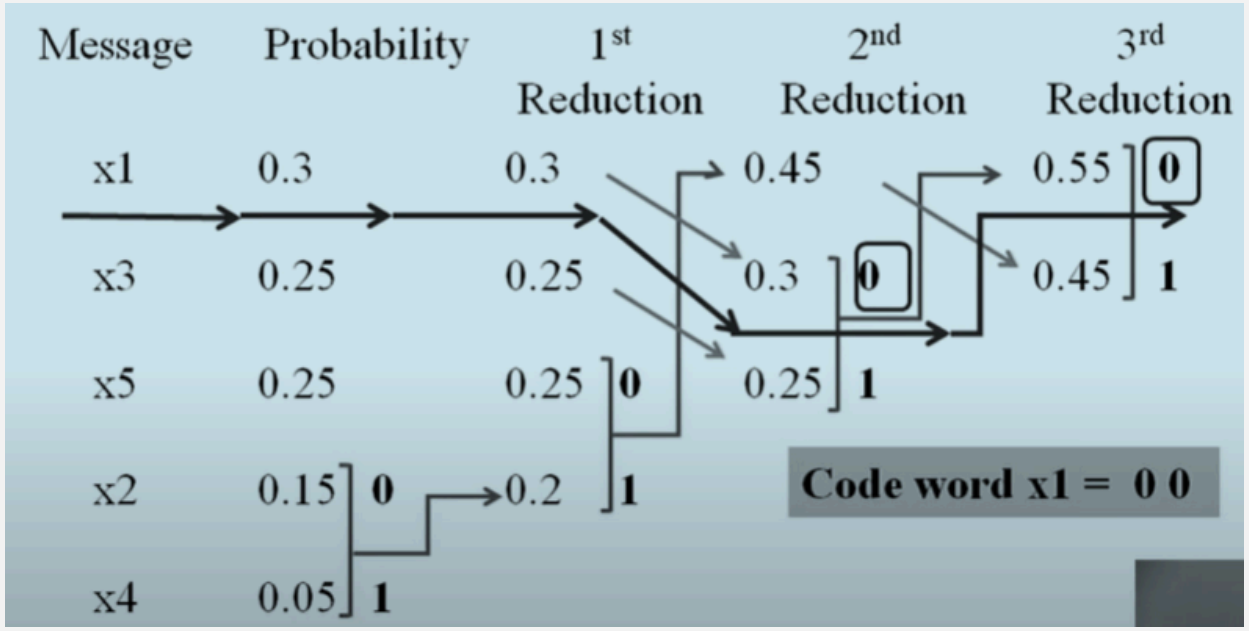


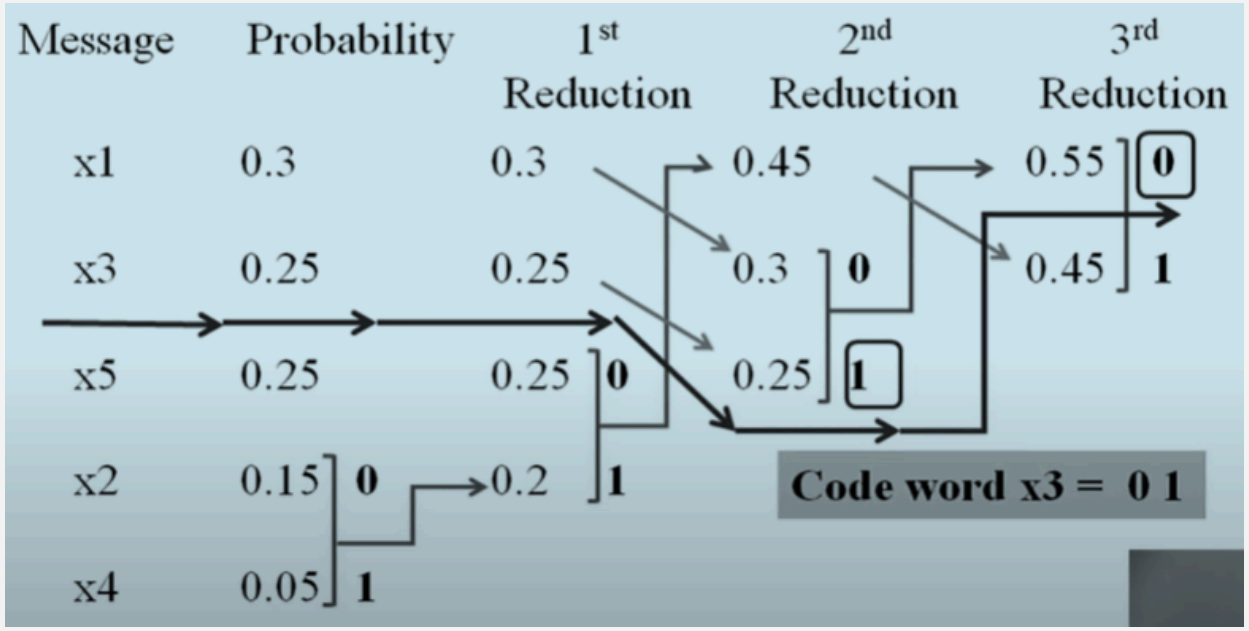
4) Repeat the steps 2 and 3 till only M probabilities remain

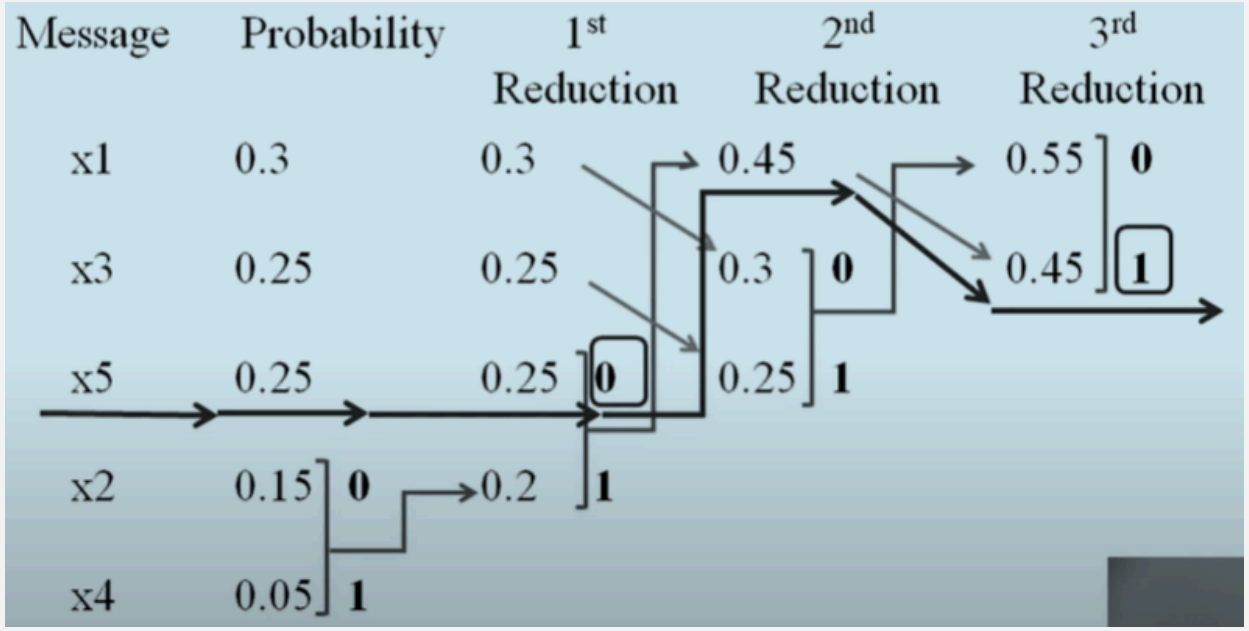


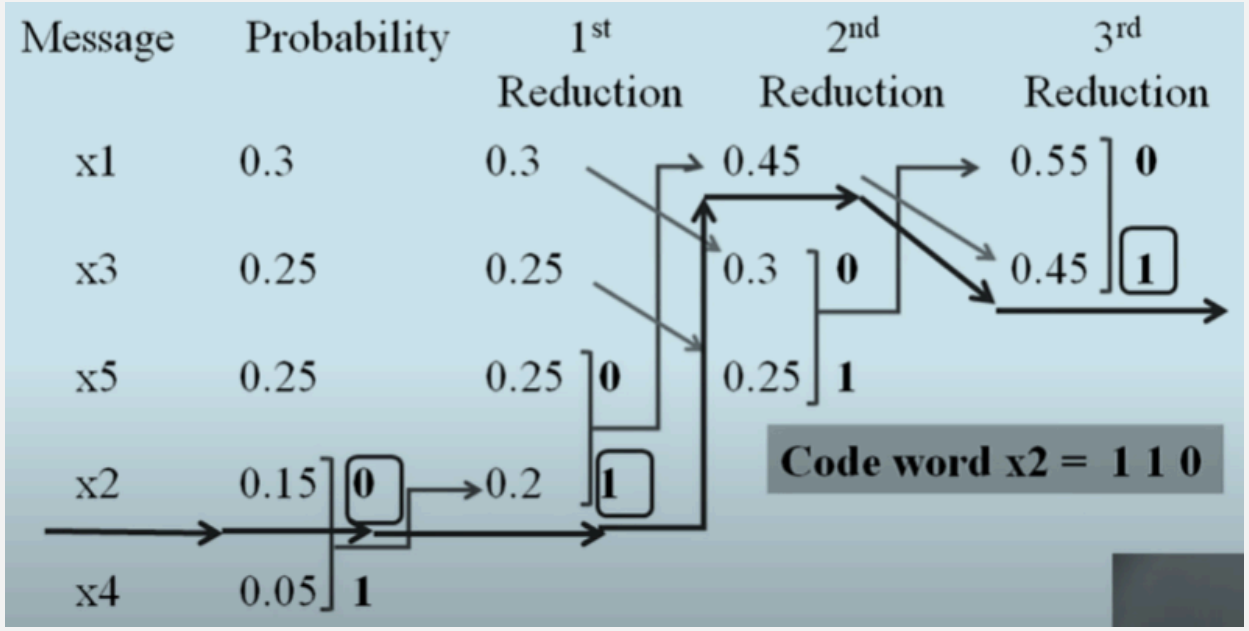
5) Obtain the code words for each message by tracing the probabilities from *left to right* in the direction of arrow till end and writing the respective symbols (0 or 1) above the path by tracing them from *right to left*.

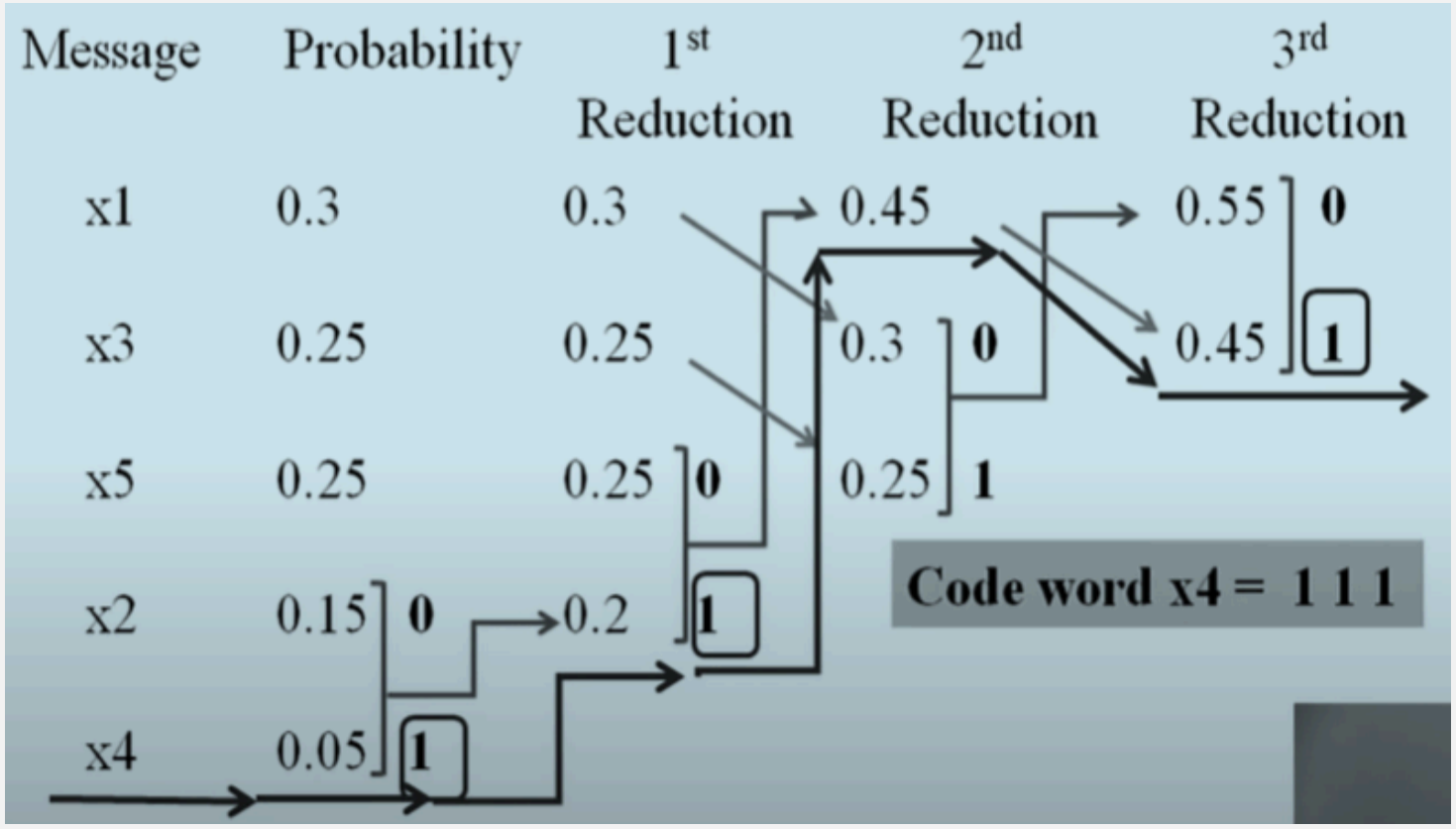



Message
Probability
1st Reduction
2nd Reduction
3rd Reduction
x1 0.3 0.3 0.45 0.55 0
x3 0.25 0.25 0.3 0 0.45 1
x5 0.25 0.25 0 0.25 1
x2 0.15 0 0.2 1
x4 0.05 1
Code word x1 = 0 0


Message
Probability
1st Reduction
2nd Reduction
3rd Reduction
x1 0.3 0.3 0.45 0.55 0
x3 0.25 0.25 0.3 0 0.45 1
x5 0.25 0.25 0 0.25 1
x2 0.15 0 0.2 1
x4 0.05 1
Code word x3 = 0 1


Message
Probability
1st Reduction
2nd Reduction
3rd Reduction
x1 0.3 0.3 0.45 0.55 0
x3 0.25 0.25 0.3 0 0.45 1
x5 0.25 0.25 0 0.25 1
x2 0.15 0 0.2 1
x4 0.05 1


Message
Probability
1st Reduction
2nd Reduction
3rd Reduction
x1 0.3 0.3 0.45 0.55 0
x3 0.25 0.25 0.3 0 0.45 1
x5 0.25 0.25 0 0.25 1
x2 0.15 0 0.2 1
x4 0.05 1
Code word x2 = 1 1 0



Message x_k	Probability p_k	Code	Length n_k
x1	0.3	00	2
x3	0.25	01	2
x5	0.25	10	2
x2	0.15	110	3
x4	0.05	111	3

Message x_k	Probability p_k	Code	Length n_k
x1	0.3	00	2
x3	0.25	01	2
x5	0.25	10	2
x2	0.15	110	3
x4	0.05	111	3

• Average Length of code word for each message

$$\bar{L} = \sum_{k=1}^{N} nk * pk$$

$$\bar{L} = n_1 * p_1 + n_2 * p_2 + n_3 * p_3 + n_4 * p_4 + n_5 * p_5$$

$$\bar{L} = 2*0.3 + 2*0.25 + 2*0.25 + 3*0.15 + 3*0.05$$

$$\bar{L} = 2.2 \text{ Letters / message}$$

Message x_k	Probability p_k	Code	Length n_k
x1	0.3	00	2
x3	0.25	01	2
x5	0.25	10	2
x2	0.15	110	3
x4	0.05	111	3

$\bar{L} = 2.2$ Letters / message

• Entropy

$$H(X) = -\sum_{k=1}^{N} pk \log p_k$$

$$H(X) = -[p_1 * \log_2 p_1 + p_2 * \log_2 p_2 + p_3 * \log_2 p_3 + p_4 * \log_2 p_4 + p_5 * \log_2 p_5]$$

$$= -[0.3\log_2 0.3 + 0.25\log_2 0.25 + 0.25\log_2 0.25 + 0.15\log_2 0.15 + 0.05\log_2 0.05]$$

$H(X) = 2.15$ Bits / message

• Efficiency

$$\eta = \frac{H(X)}{\bar{L} * Log_2 M}$$

$$= 2.15 / (2.2 * \text{Log}_2 2)$$

$$= 0.977$$

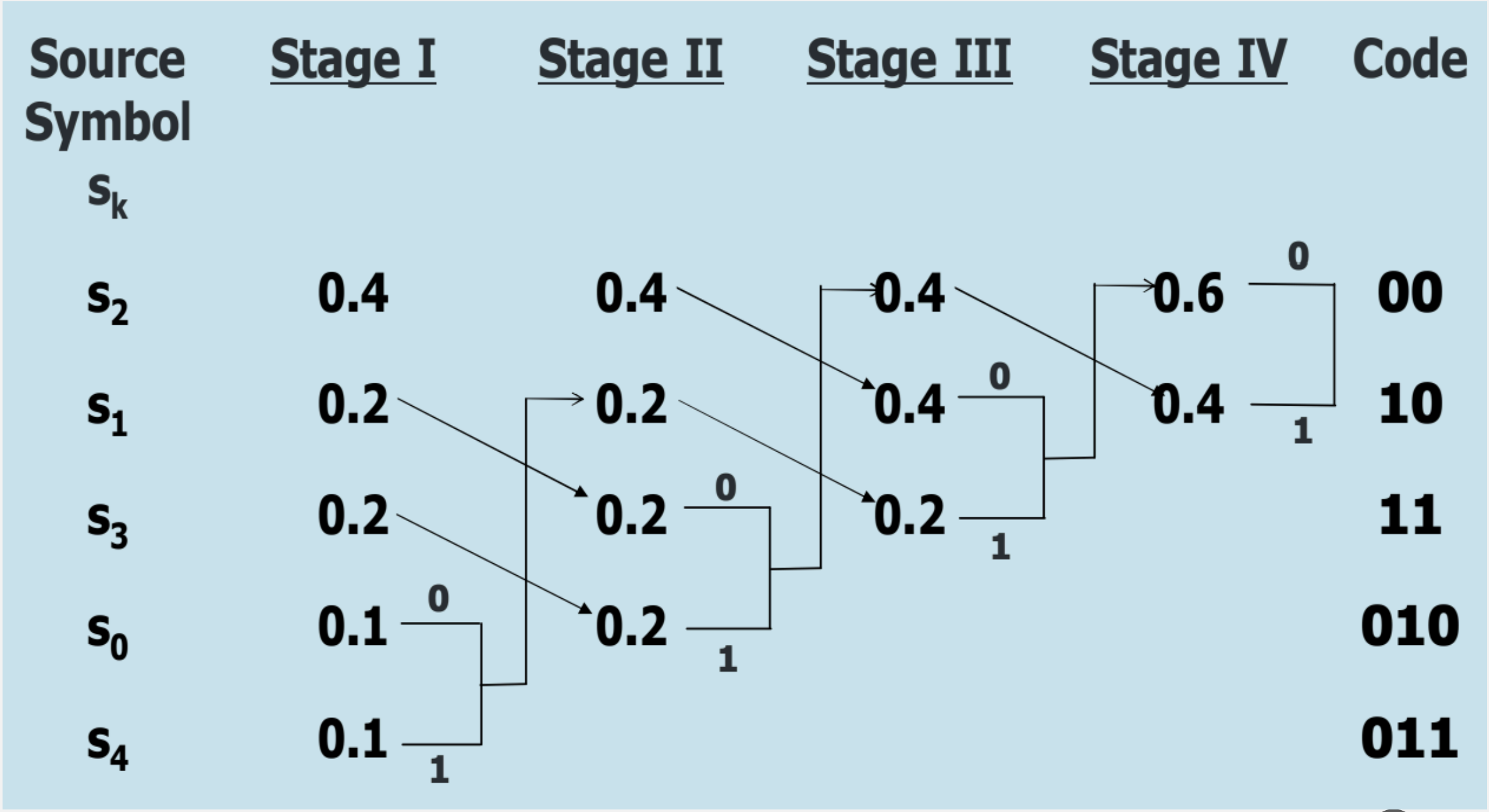
$$\eta = 97.7\ \%$$

Problem Statement

A DMS has an alphabet of five symbols whose probabilities of occurrence are as described here

1. Compute the Huffman Code for this source.
2. Find the average code word length and entropy of this code.
3. On what percentage the average code word length does exceed the entropy.

Source Symbol s_k	Symbol Probability p_k
s_0	0.1
s_1	0.2
s_2	0.4
s_3	0.2
s_4	0.1


Source Symbol
Stage I
Stage II
Stage III
Stage IV
Code
s_k
s_2
s_1
s_3
s_0
s_4
0.4
0.2
0.2
0.1
0.1
0.4
0.2
0.2
0.4
0.4
0.2
0.6
0.4
00
10
11
010
011
0
1

(ii) W.K,
the average code word length,

$$L = \sum_{k=0}^{k=4} P_k l_k$$

$$= 0.4 \times (2) + 0.2 \times (2) + 0.2 \times (2) + 0.1 \times (3) + 0.1 \times (3)$$

$$= 2.2 \text{ (Ans)}$$

And also know that

Entropy, $H(S) = \sum_{P_k} P_k \log_2 1/P_k$

$$= 0.4 \log_2 \frac{1}{0.4} + 0.2 \log_2 \frac{1}{0.2} + 0.2 \log_2 \frac{1}{0.2} + 0.1 \log_2 \frac{1}{0.1} + 0.1 \log_2$$

$$= 2.12 \text{ (Ans)}$$

(iii) The variance of L is

$$\sigma^2 = \sum_{k=0}^{4} P_k (l_k - \bar{L})^2$$

$$= 0.4(2-2.2)^2 + 0.2(2-2.2)^2 + 0.2(2-2.2)^2 + 0.1(3-2.2)^2 + 0.1(3-2.2)$$

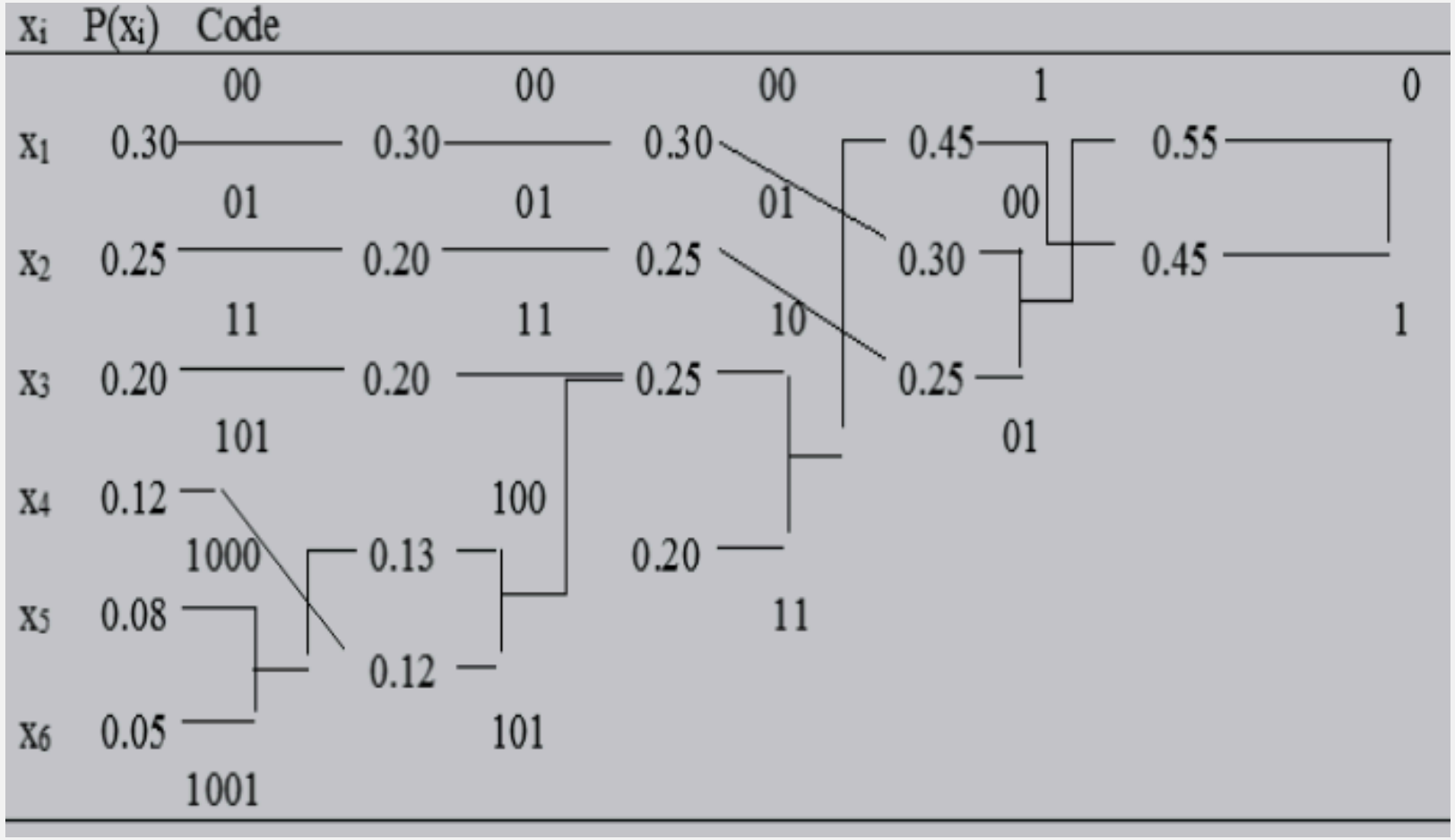
$$= 0.16 \text{ (Ans)}$$

Problem Statement

A DMS has an alphabet of five symbols whose probabilities of occurrence are as described here

1. Compute the Huffman Code for this source.
2. Find the average code word length and entropy of this code.
3. On what percentage the average code word length does exceed the entropy.

Source Sample x_i	$P(x_i)$
X_1	0.30
X_2	0.25
X_3	0.20
X_4	0.12
x_5	0.08
x_6	0.05



Source Sample x_i	$P(x_i)$	Codeword
X_1	0.30	00
X_2	0.25	01
X_3	0.20	11
X_4	0.12	101
x_5	0.08	1000
x_6	0.05	1001

$H(X) = 2.36$ b/symbol

$\bar{L} = 2.38$ b/symbol

$\eta = H(X)/\bar{L} = 0.99$

Problem Statement:

A file contains the following characters with the frequencies as shown. If Huffman Coding is used for data compression, determine-

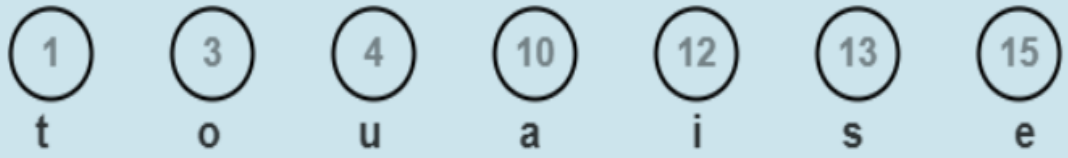
1. Huffman Code for each character
2. Average code length
3. Length of Huffman encoded message (in bits)

Characters	Frequencies
a	10
e	15
i	12
o	3
u	4
s	13
t	1

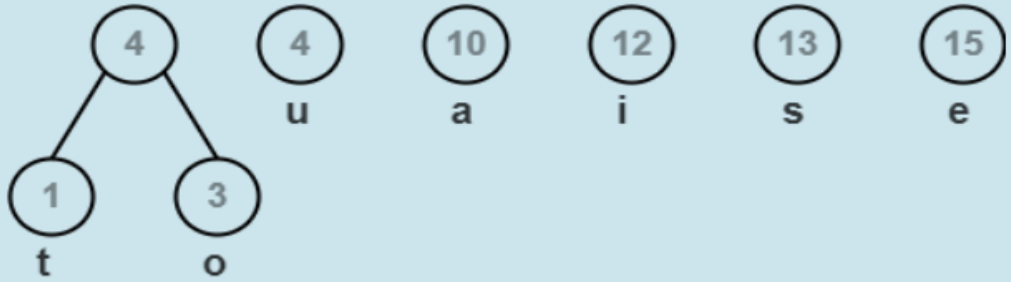
Characters	Frequencies
a	10
e	15
i	12
o	3
u	4
s	13
t	1

It follows **Optimal Most Pattern**

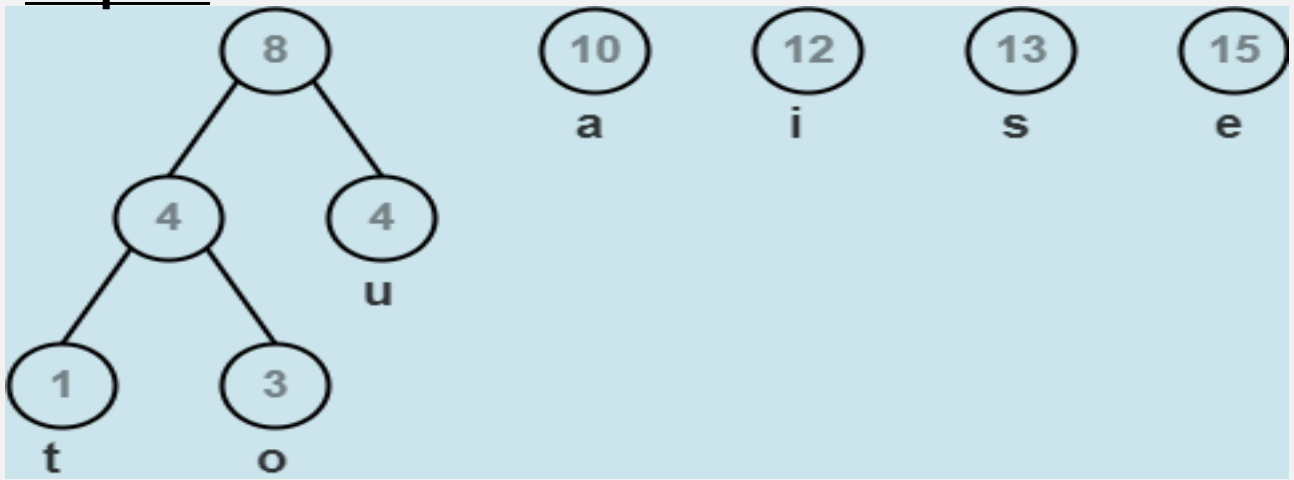
Step-01:



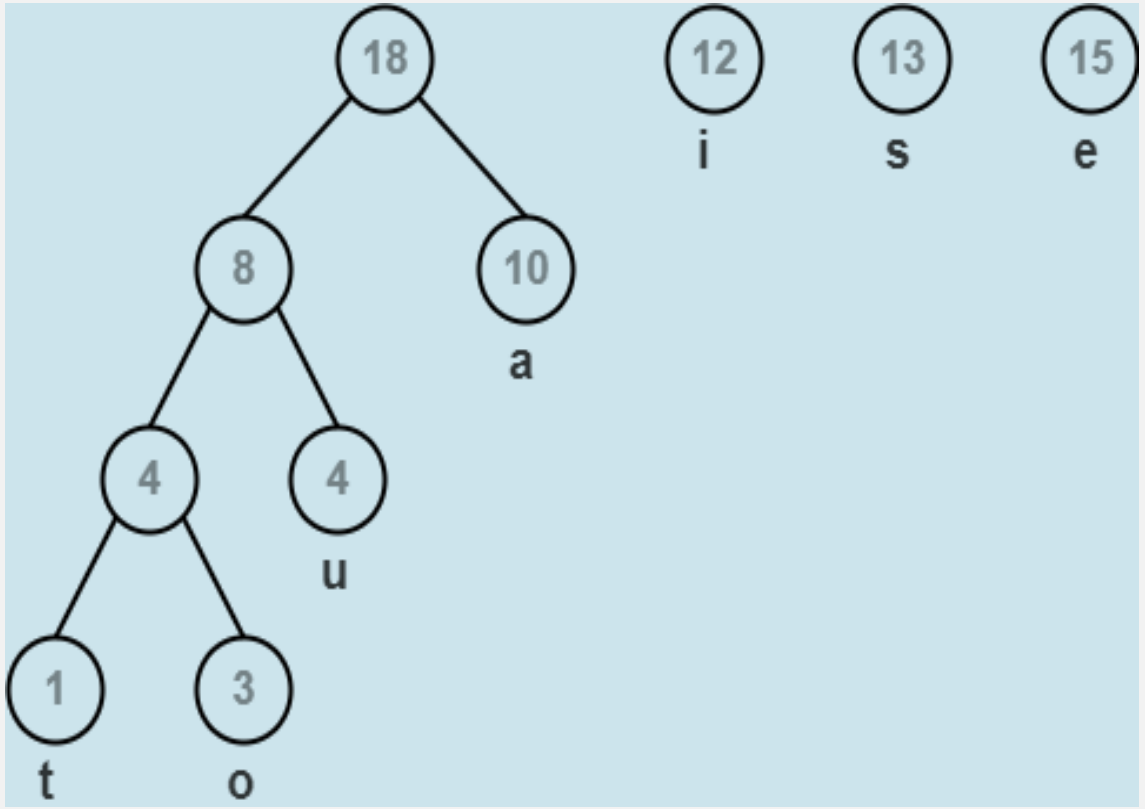
Step-02:



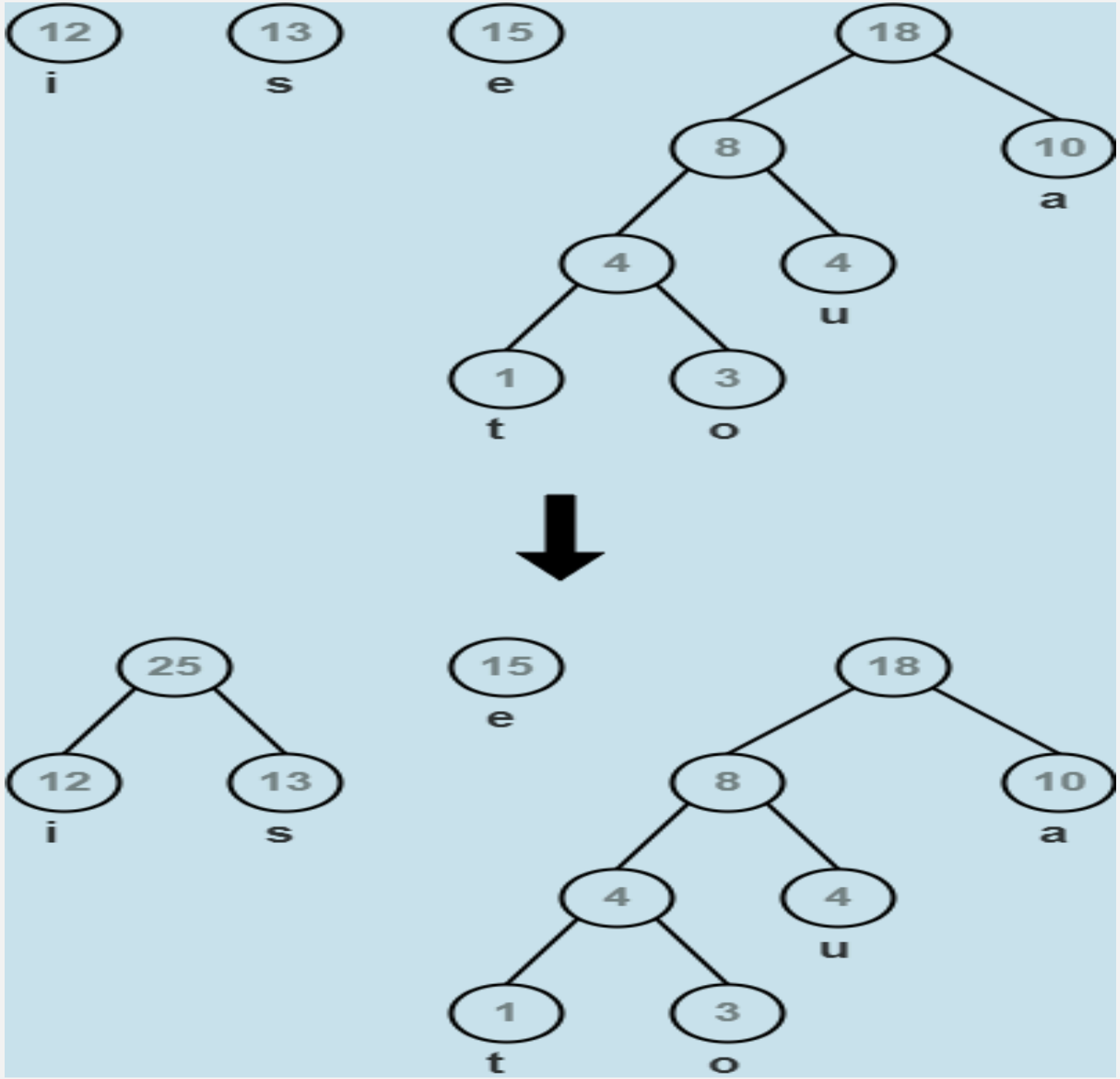
Step-03:



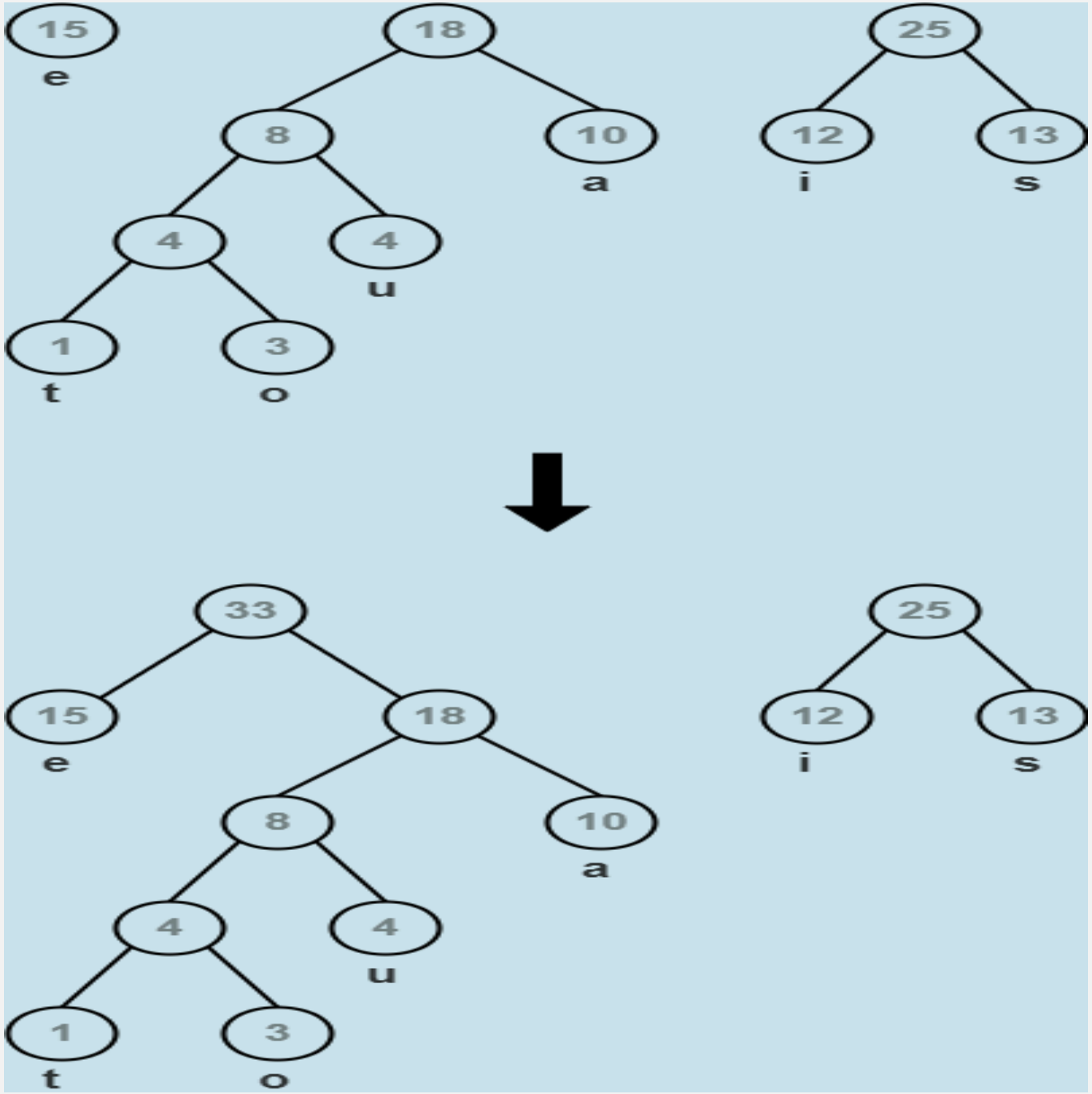
Step-04:



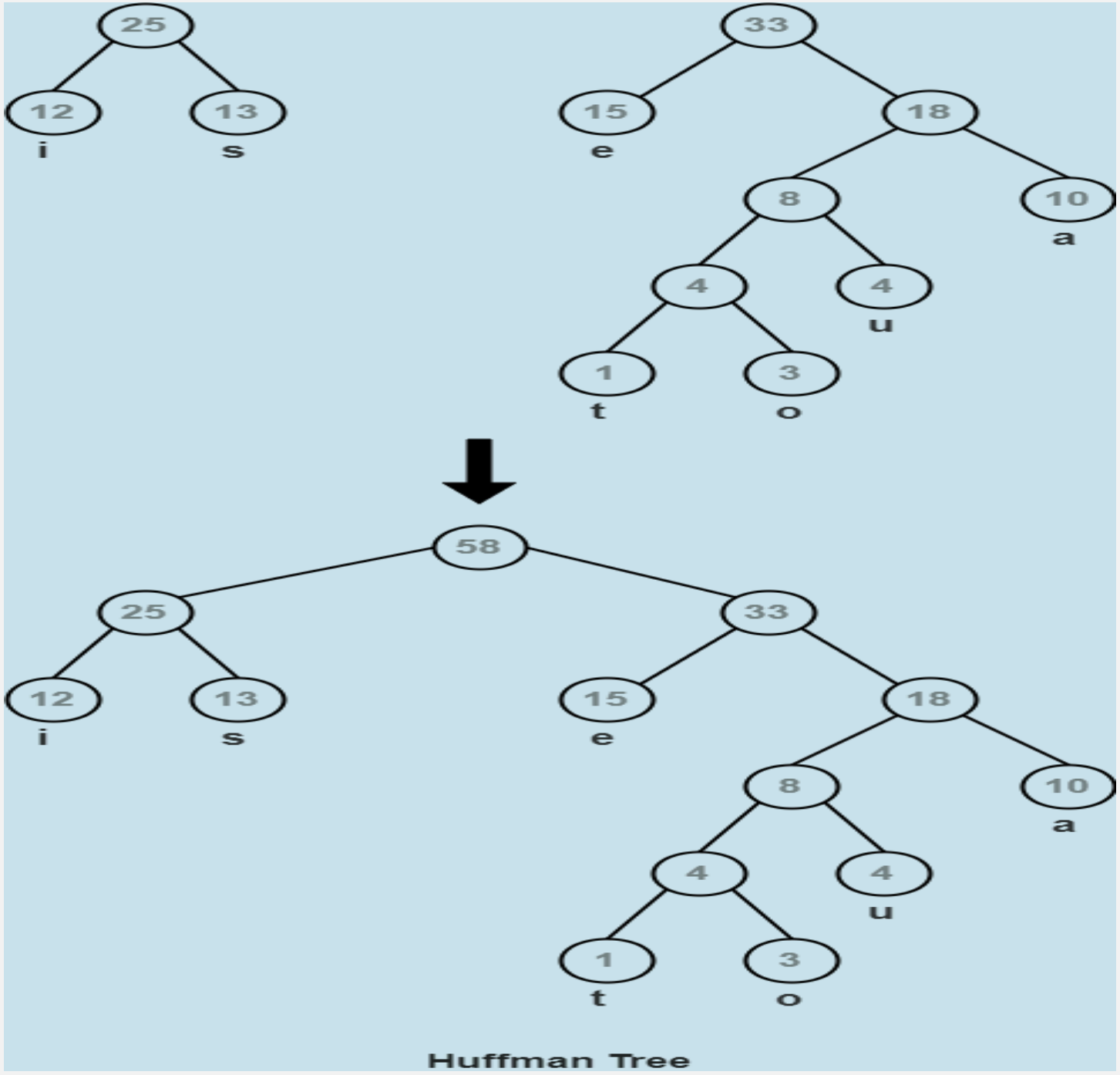
Step-05:



Step-06:

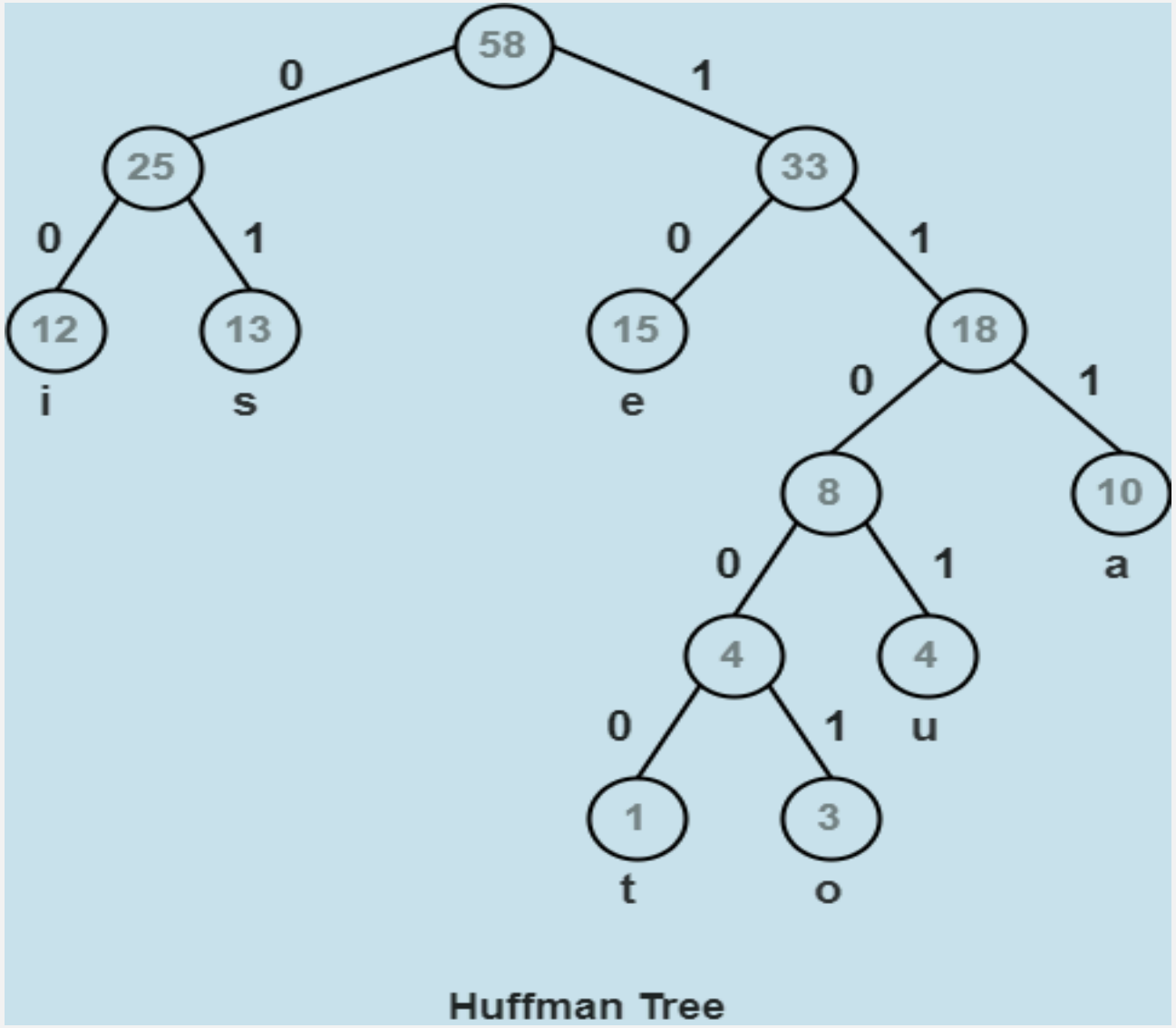

15
e
18
8
10
a
4
4
u
1
t
3
o
25
12
i
13
s
33
15
e
18
8
10
a
4
4
u
1
t
3
o
25
12
i
13
s

Step-07:



Huffman Tree

After assigning weight to all the edges, the modified Huffman Tree is-



Huffman Tree

Following this rule, the Huffman Code for each character is-

- a = 111
- e = 10
- i = 00
- o = 11001
- u = 1101
- s = 01
- t = 11000

Average Code Length-

Average code length

$= \sum (\text{frequency}_i \times \text{code length}_i) / \sum (\text{frequency}_i)$

$= \{ (10 \times 3) + (15 \times 2) + (12 \times 2) + (3 \times 5) + (4 \times 4) + (13 \times 2) + (1 \times 5) \} / (10 + 15 + 12 + 3 + 4 + 13 + 1)$

Length of Huffman Encoded Message-

Total number of bits in Huffman encoded message

= Total number of characters in the message x Average code length per character

= 58 x 2.52

= 146.16

≅ 147 bits

Problem Statement:

A file contains the following characters with the frequencies as shown. If Huffman Coding is used for data compression, determine-

1. Huffman Code for each character
2. Average code length
3. Length of Huffman encoded message (in bits)

SYMBOL	PROBABILITY
A	.36
B	.15
C	.13
D	.11
E	.09
F	.07
G	.05
H	.03
I	.01

9/25
3/20
13/100
11/100
9/100
7/100
3/100
1/100

LECTURE – 7

Prefix Code

Prefix-free code and prefix code are actually the same things (forget about how the namings were done). The formal definition of a prefix-free code system from Wikipedia is as follows:

"A prefix code is a type of code system (typically a variable-length code) distinguished by its possession of the "prefix property", which requires that there is no whole code word in the system that is a prefix (initial segment) of any other code word in the system. "

Concretely, let us see an example. We have the following one-to-one binary code system, represented using a dictionary.

- A **prefix code** is a type of code in which **no codeword is a prefix of any other codeword**.
- This property ensures that the encoded message can be uniquely decoded without ambiguity.
- Example: {**0, 10, 110, 111**} → Here, no codeword is the starting part (prefix) of another.

A: 00
B: 010
C: 011
D: 10
E: 11

We could check the word one by one to see whether the system satisfies the "prefix property". A was coded as 00, while none of the codes of B, C, D, and E are started with 00. Using the same way, we check B, C, D, and E, and confirm that it is a prefix-free code system.

If we change the above system a little bit to the following system:

A: 00
B: 010
C: 001 # 011 -> 001
D: 10
E: 11

The new system is still one-to-one correspondence. However, it is no-longer a prefix-free code system, because the code of A 00 was shown as the prefix in the code of C which is 001.

Example of Prefix Code:

Source Symbol	Probability	Code I	Code II	Code III
s0	0.5	0	0	0
s1	0.25	1	10	01
s2	0.125	00	110	011
s3	0.125	11	111	0111

- ***Code I*** : its not a prefix code since the bit 0, the codeword for s0, is a prefix of 00 (s2). Again the bit 1, the codeword for s1 is the prefix of 11 (s3).

- ***Code III***: its also not a prefix code. Justify…

- ***Code II***: it's a prefix code for the given condition that no code is the prefix of any other.

Examples

X	Code 1	Code 2	Code 3	Code 4
1	01	0	10	0
2	110	010	00	10
3	010	10	11	110
4	110	01	110	111

x_i	Code 1	Code 2	Code 3	Code 4	Code 5	Code 6
x_1	00	00	0	0	0	1
x_2	01	01	1	10	01	01
x_3	00	10	00	110	011	001
x_4	11	11	11	111	0111	0001

LEMPEL-ZIV CODING

Data compression, also called compaction, the process of reducing the amount of data needed for the storage or transmission of a given piece of information, typically by the use of encoding techniques. Compression predates digital technology, having been used in Morse Code, which assigned the shortest codes to the most common characters, and in telephony, which cuts off high frequencies in voice transmission. Today, when an uncompressed digital image may require 20 megabytes, data compression is important in storing information digitally on computer disks and in transmitting it over communications networks.

Data compression may be lossless (exact) or lossy (inexact). Lossless compression can be reversed to yield the original data, while lossy compression loses detail or introduces small errors upon reversal. Lossless compression is necessary for text, where every character is important, while lossy compression may be acceptable for images or voice (the limitation of the frequency spectrum in telephony being an example of lossy compression). The three most common compression programs for general data are Zip (on computers using Windows operating system), StuffIt (on Apple computers), and gzip (on computers running UNIX); all use lossless compression. A common format for compressing static images, especially for display over the Internet, is GIF (graphics interchange format), which is also lossless except that its images are limited to 256 colors. A greater range of colors can be used with the JPEG (joint photographic experts group) formatting standard, which uses both lossless and lossy techniques, as do various standards of MPEG (moving picture expert group) for videos.

Adaptive dictionary methods build a table of strings and then replace occurrences of them by shorter codes. The Lempel-Ziv algorithm, invented by Israeli computer scientists Abraham Lempel and Jacob Ziv, uses the text itself as the dictionary, replacing later occurrences of a string by numbers indicating where it occurred before and its length. Zip and gzip use variations of the Lempel-Ziv algorithm.

- Lempel-Ziv-Welch (LZW) coding is the foremost technique for general purpose data compression due to its simplicity and versatility.
- It is invented by Abraham Lempel, Jacob Ziv, and Terry Welch.
- It addresses spatial redundancies in an image.
- It is an error free compression approach.
- Prior knowledge of probability of occurrence of symbols to be encoded is not required.
- Lossless compression technique

Why LZW coding?

- For most palette color images LZW yields the highest compression efficiency without sacrificing image data.
- The GIF image file format is the de facto standard for images on the web. Most GIF files use the LZW compression.

Algorithm for Arithmetic Coding

The steps of the encoding algorithm are as follows:

Step-1: Consider a range of real numbers in [0,1]. Subdivide this range into a number of sub-ranges that is equal to the total number of symbols in the source alphabet. Each sub-range spans a real value equal to the probability of the source symbol.

Step-2: Consider a source message and take its first symbol. Find to which subrange does this source symbol belongs.

Step-3: Subdivide this sub-range into a number of next level sub-ranges, according to the probability of the source symbols.

Step-4: Now parse the next symbol in the given source message and determine the next-level sub-range to which it belongs.

Step-5: Repeat step-3 and step-4 until all the symbols in the source message are parsed. The message may be encoded using any real value in the last sub-range so formed. The final message symbol is reserved as a special end-of-symbol message indicator.

Lempel-Ziv Coding Example

0 0 0 1 0 1 1 1 0 0 1 0 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1							
Representation									
Encoding									

00 1 0 1 1 1 0 0 1 0 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00						

0 0 0 1 0 1 1 1 0 1 0 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011	10			

00 01 1 1 1 0 0 1 0 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01					

0 0 0 1 0 1 1 1 0 010 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011	10	010		

0 0 0 1 011 1 0 0 1 0 1 0 0 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011				

0 0 0 1 0 1 1 1 0 0 1 0 100 1 0 1...

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011	10	010	100	

0 0 0 1 0 1 1 1 0 0 1 0 1 0 0 1 0 1..

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011	10	010	100	101
Representation									
Encoding									

Information bits	0 0	0 1	0 1 1	1 0	0 1 0	1 0 0	1 0 1	...	
Source encoded bits	0010	0011	1001	0100	1000	1100	1101		

Codebook Index	1	2	3	4	5	6	7	8	9
Subsequence	0	1	00	01	011	10	010	100	101
Representation			11	12	42	21	41	61	62
Source Code			0010	0011	1001	0100	1000	1100	1101

Example:

AABABBBABAABABBBABBABB

Example:

AABABBBABAABABBBABBABB

A → 0

B → 1

Position	1	2	3	4	5	6	7	8	9
Sequence									
Representation									
Encoding									

Example:

AABABBBABAABABBBABBABB

A → 0

B → 1

A-AB-ABB-B-ABA-ABAB-BB-ABBA-BB

Position	1	2	3	4	5	6	7	8	9
Sequence	A	AB	ABB	B	ABA	ABAB	BB	ABBA	BB
Representation	ϕ A	1 B	2 B	ϕ B	2 A	5 B	4 B	3 A	7
Encoding	0	1 1	10 1	1	10 0	101 1	100 1	11 0	111
	0000	0011	0101	0001	0100	1011	1001	0110	0111

Example:

ABAABABABBABABABABBABBAAB

ABAABABABBABABABABBABBAAB

A-B-AA-BA-BAB-BABA-BABAB-BABB-AAB

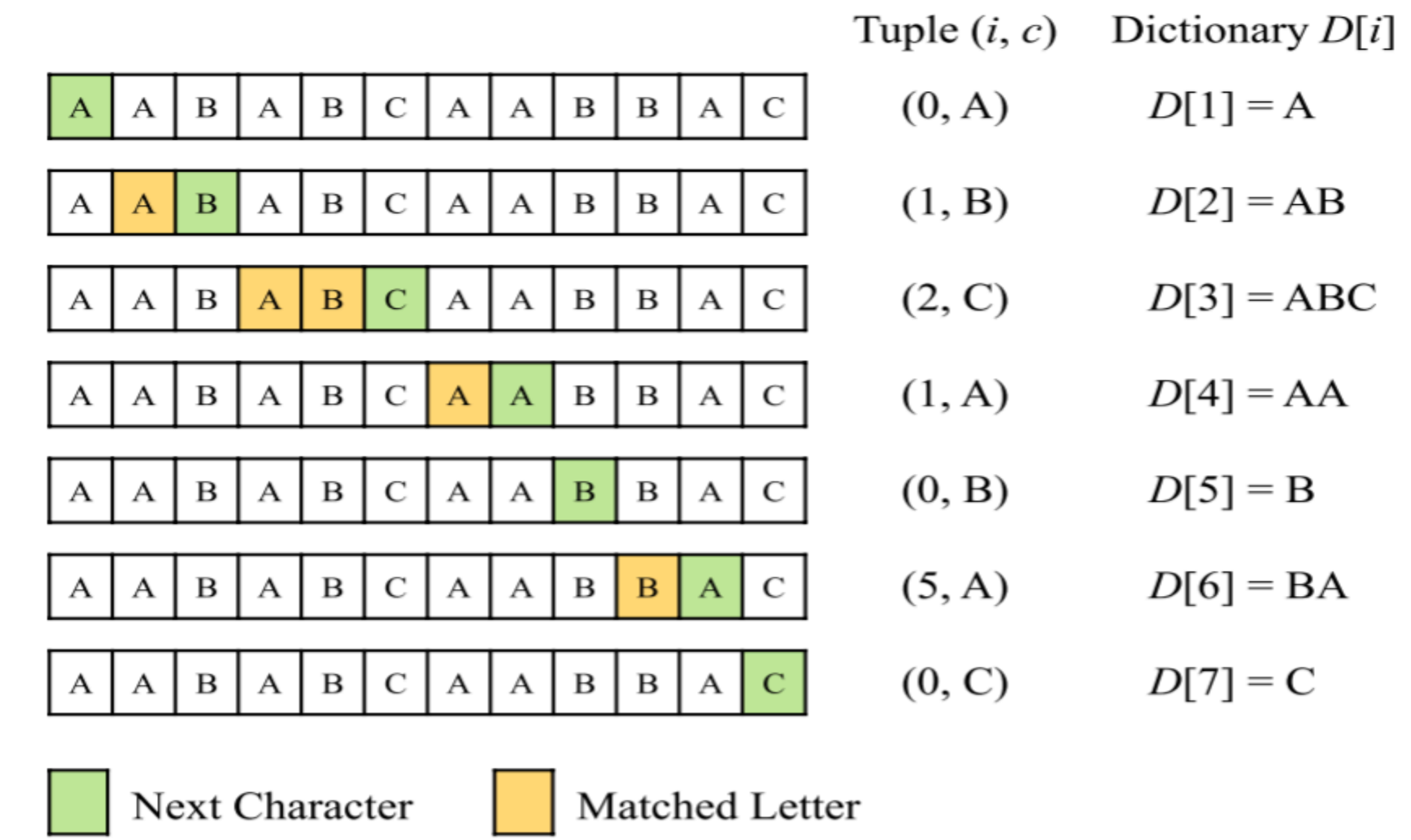
Position	1	2	3	4	5	6	7	8	9
Sequence	A	B	AA	BA	BAB	BABA	BABAB	BABB	AAB
Representation	ϕ A	ϕ B	1 A	2 A	4 B	5 A	6 B	5 B	3 B
Encoding	0	1	1 0	10 0	100 1	101 0	110 1	101 1	111
	0000	0001	0010	0100	1001	1010	1101	1011	0111

ABAABABABBABABABABBABBAABBABABAAB

A-B-AA-BA-BAB-BABA-BABAB-BABB-AAB-BABABA-AB

Position	1	2	3	4	5	6	7	8	9	10	11
Sequence	A	B	AA	BA	BAB	BABA	BABAB	BABB	AAB	BABABA	AB
Representation	ϕ A	ϕ B	1 A	2 A	4 B	5 A	6 B	5 B	3 B	7 A	1 B
Encoding	0	1	1 0	10 0	100 1	101 0	110 1	101 1	111	111 0	1 1
	0000	0001	0010	0100	1001	1010	1101	1011	0111	1110	0011

Example of compression process of LZ78 for the text data "AABABCAABBAC"



Data Compression

- Compression is a reduction in the number of bits needed to represent data.
- Compressing data can save storage capacity, speed file transfer, and decrease costs for storage hardware and network bandwidth.
- Compression is performed by a program that uses a formula or algorithm to determine how to shrink the size of the data.

Data Compression refers to a technique where a large file to reduced to smaller sized file and can be decompressed again to the large file.

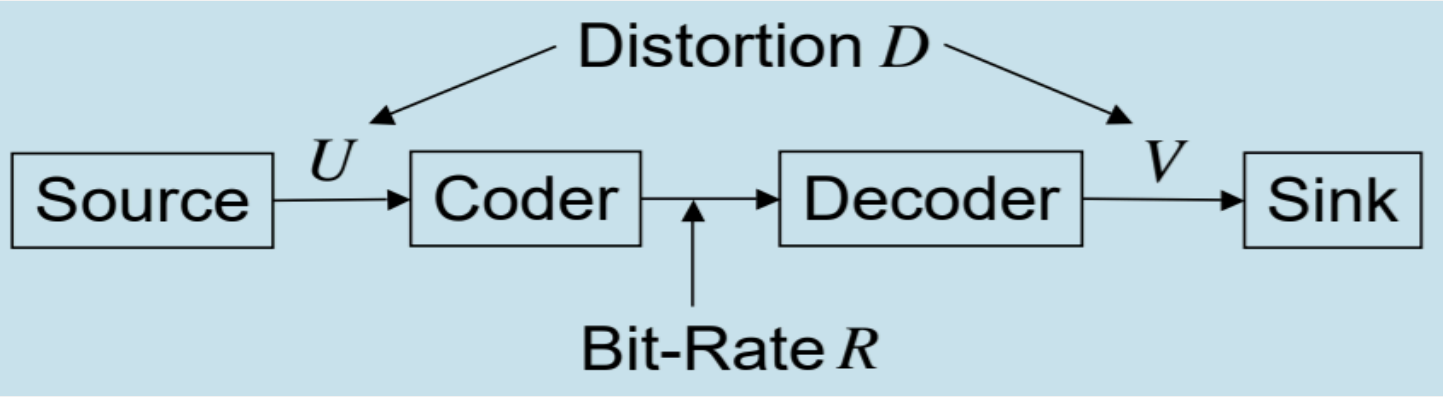
Data Compression is a technique in which the size of data is reduced without loss of information.

Lossy compression and **Lossless compression** are the categories of data compression method.

The **lossy compression** technique does not restored the data in its original form, after decompression on the other hand **lossless compression** restores and rebuilt the data in its

Rate–distortion Theory

Rate–distortion theory is a major branch of information theory which provides the theoretical foundations for lossy data compression; it addresses the problem of determining the minimal number of bits per symbol, as measured by the rate R, that should be communicated over a channel, so that the source (input signal) can be approximately reconstructed at the receiver (output signal) without exceeding an expected distortion D.



Rate distortion theory calculates the minimum transmission bit-rate R for a required picture quality.

Results of rate distortion theory are obtained without consideration of a specific coding method.

- Symbol *u* sent, *v* received
- Per-symbol distortion:

$$d(u,v) \geq 0$$

$$d(u,v) = 0 \quad \text{for } u = v$$

- Average distortion:

$$D(U,V) = E\{d(U,V)\} = \sum_u \sum_v P(u,v)d(u,v)$$

- Distortion criterion:

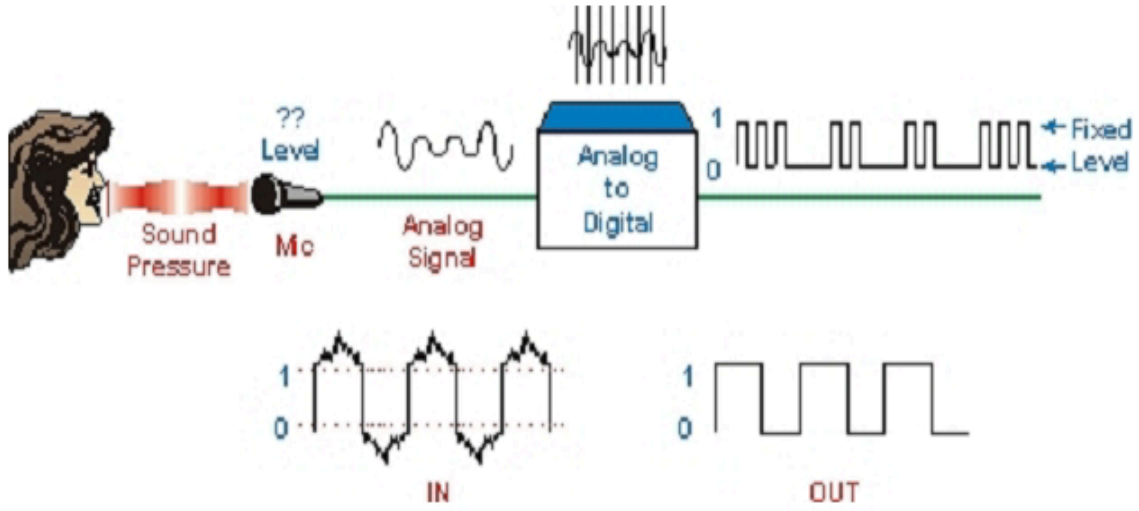
$$D \leq D^*$$

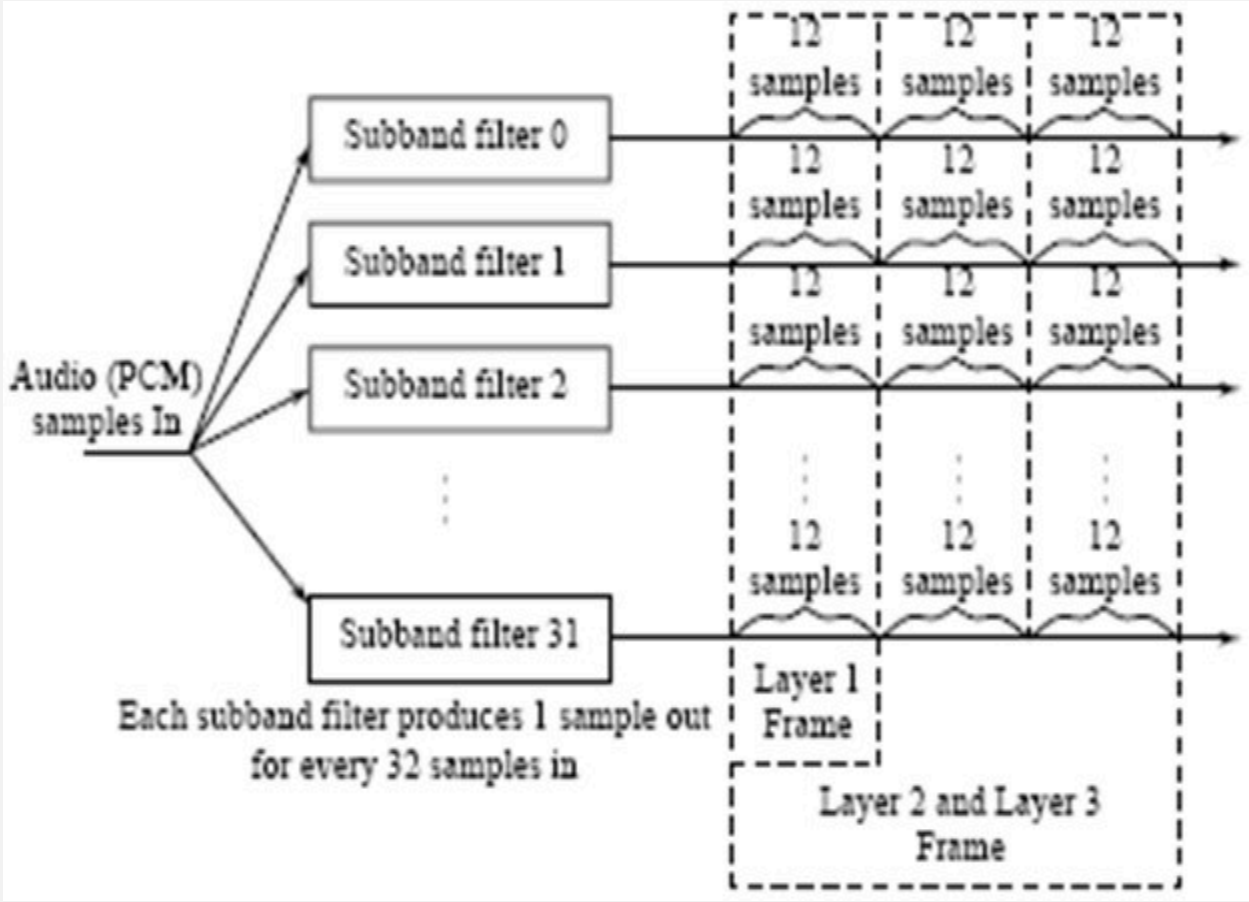
D^*: maximum average distortion

MPEG Audio Compression

- MPEG stands for MOVING PICTURE EXPERTS GROUP is a standard for video and audio compression for eliminating the noisy signals from the transmitted signals from the satellite.
- It sets the standard for audio and video compression and its transmission.
- Today in the field of communication compression standards from MPEG-1 to MPEG DASH have been developed that give digital output.
- MPEG is a first international algorithm for digital audio compression that showed high fidelity.
- Data compression is a technique in which data content of the input signal to system is compressed so that original signal is filtered out and unwanted or undesired signals are removed and true digital signal is obtained as output.
- Therefore when audio signals are used in the form of data it is termed as AUDIO COMPRESSION.
- In this compression technique reducing the bit rate which is required is an important task for perfect design of an audio system.
- High quality audio transmission and storage are the basic goals of audio compression.
- The MPEG is a generic audio compression standard utilizing technology which compresses audio signal to make it audible to the human ear and remove the unwanted signals introduces in the communication path without assuming nature of audio source.

- Digitization
- Compression
- Multiple Channels





Audio Types

- Level 1 - Low Complexity
- Level 2 - MUSICAM (DVB)
- Level 3 - MP3
- Advanced Audio Coding

MPEG layer 1 audio is a low complexity audio compression system. MP1 was the first digital audio and it uses the precision adaptive sub-band coding (PASC) algorithm. This process divides the digital audio signal into multiple frequency bands and only transmits the audio bands that can be heard by the listener. To obtain high fidelity quality (e.g. music) with MP1 typically requires 192 kbps per audio channel.

MPEG layer 2 audio is a medium complexity audio compression system which is also known as the MUSICAM system. The MUSICAM system achieves medium compression ratios dividing the audio signal into sub bands, coding these sub bands and multiplexing them together. The MUSICAM system is used in the digital audio broadcasting (DAB) system. To obtain high fidelity quality (e.g. music) with MP2 typically requires 128 kbps per audio channel.

MPEG layer 3 is a lossy audio coding standard that uses a relatively high-complexity audio analysis system to characterize and highly compress audio signals. The MP3 system achieves high-compression ratios (10:1 or more) by removing redundant information and sounds that the human ear cannot detect or perceive. The removal of information components that cannot be detected (such as low level signals that occur simultaneously with high-level signals) is called psychoacoustic compression. To obtain high fidelity quality (e.g. music) with MP3 typically requires 64 kbps per audio channel.

VIDEO COMPRESSION

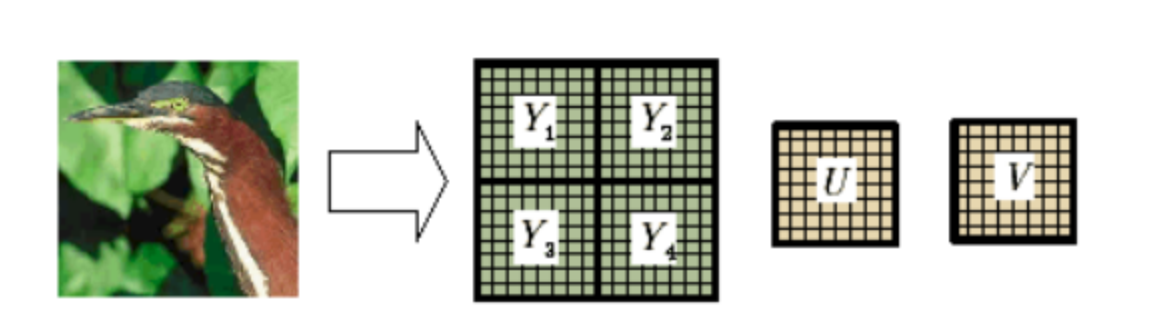
- Video compression uses modern coding techniques to reduce redundancy in video data.
- Video takes up a lot of space. Uncompressed footage from a camcorder takes up about 17MB per second of video.
- Because it takes up so much space, video must be compressed before it is put on the web. “Compressed” just means that the information is packed into a smaller space.
- There are two kinds of compression:
  - Lossy Video Compression
  - Lossless Video Compression

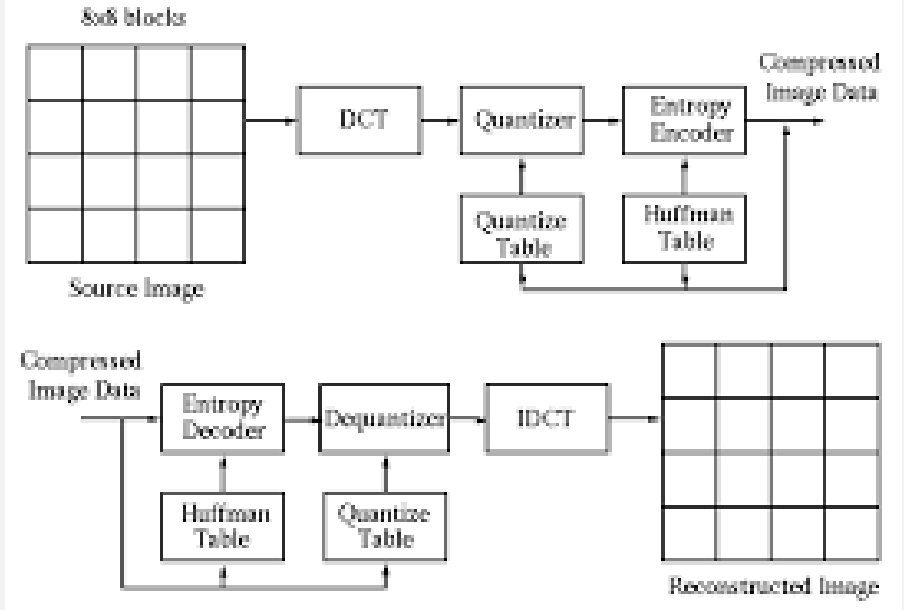
- MPEG **algorithms** compress data to form small bits that can be easily transmitted and then decompressed. MPEG achieves its high compression rate by storing only the changes from one frame to another, instead of each entire frame. The video information is then encoded using a technique called **Discrete Cosine Transform** (DCT). MPEG uses a type of lossy compression, since some data is removed. But the diminishment of data is generally imperceptible to the human eye.

JPEG Image Compression Standard

JPEG is an image compression standard which was developed by "Joint Photographic Experts Group". In 1992, it was accepted as an international standard. JPEG is a lossy image compression method. JPEG compression uses the DCT (Discrete Cosine Transform) method for coding transformation. It allows a tradeoff between storage size and the degree of compression can be adjusted.

JPEG compression is mainly discards much of the original information so the exact original image cannot be reconstructed from a JPEG file. This is called lossy compression. While this sounds bad, a photograph actually contains considerable information that the human eye cannot detect so this can be safely discarded.





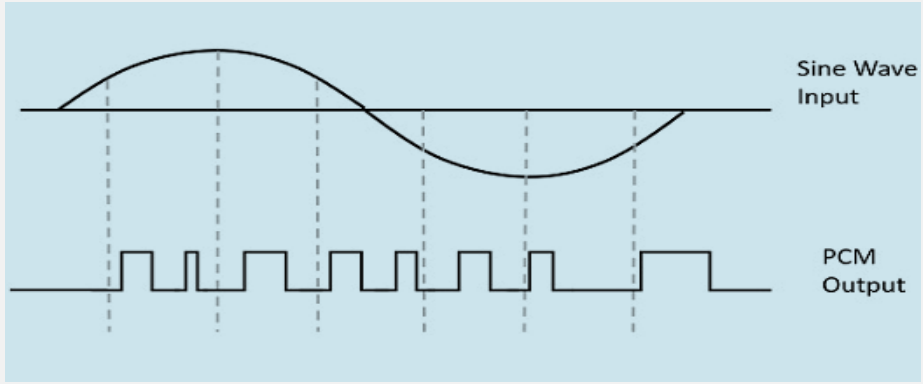
Waveform Coding

Waveform coding is the type of speech digitization used in the public switched (wireline) telephony network.

Waveform coding techniques describe the waveform' s instantaneous behavior.

The world's communication systems have converted to a digital transmission format called pulse code modulation (PCM).

PCM is a type of coding that is called "waveform" coding because it creates a coded form of the original voice waveform.



Waveform coding procedures transform a waveform set (representing a message set) into an improved waveform set. The improved waveform set can then be used to provide improved P_B compared to the original set. The most popular of such *waveform codes* are referred to as *orthogonal* and *biorthogonal codes*. The

6.1.3.1 Orthogonal Codes

A one-bit data set can be transformed, using *orthogonal codewords* of two digits each, described by the rows of matrix $\mathbf{H}_1$ as follows:

Data set	Orthogonal codeword set
0	$\mathbf{H}_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (6.4a)
1	

For this, and the following examples, use Equation (6.3) to verify the orthogonality of the codeword set. To encode a 2-bit data set, we extend the foregoing set both horizontally and vertically, creating matrix $\mathbf{H}_2$.

Data set	Orthogonal codeword set
0 0	$\mathbf{H}_2 = \left[\begin{array}{cc:cc} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ \hdashline 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right] = \begin{bmatrix} \mathbf{H}_1 & \mathbf{H}_1 \\ \mathbf{H}_1 & \overline{\mathbf{H}_1} \end{bmatrix}$ (6.4b)
0 1	
1 0	
1 1	

The lower right quadrant is the complement of the prior codeword set. We continue the same construction rule to obtain an orthogonal set $\mathbf{H_3}$ for a 3-bit data set.

Data Set
0 0 0
0 0 1
0 1 0
0 1 1
1 0 0
1 0 1
1 1 0
1 1 1

$$\mathbf{H}_3 = \left[\begin{array}{cccc:cccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] = \begin{bmatrix} \mathbf{H}_2 & \mathbf{H}_2 \\ \mathbf{H}_2 & \overline{\mathbf{H}_2} \end{bmatrix} \quad (6.4c)$$

Orthogonal codeword set

In general, we can construct a codeword set $\mathbf{H}_k$, of dimension $2^k \times 2^k$, called a *Hadamard matrix,* for a k-bit data set from the $\mathbf{H}_{k-1}$ matrix, as follows:

$$\mathbf{H}_k = \begin{bmatrix} \mathbf{H}_{k-1} & \mathbf{H}_{k-1} \\ \mathbf{H}_{k-1} & \overline{\mathbf{H}_{k-1}} \end{bmatrix} \tag{6.4d}$$

Each pair of words in each codeword set $\mathbf{H}_1, \mathbf{H}_2, \mathbf{H}_3, \ldots, \mathbf{H}_k, \ldots$ has as many digit agreements as disagreements [2]. Hence, in accordance with Equation (6.3), $z_{ij} = 0$ (for $i \neq j$), and each of the sets is orthogonal.